# Generating biophysical-simulator parameters from observed behaviour alone: a general route past the invasive-measurement bottleneck, demonstrated across worm and fly

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**Plain-language summary.** Whole-organism computer simulators of the worm *C. elegans* and the fly *Drosophila* let researchers test how brains and bodies produce behaviour. But each simulator carries thousands of internal numbers (parameters) that normally have to be tuned with invasive electrical recordings that most behaviour laboratories cannot do. We show that a laboratory only needs ordinary tracking video of an animal: a procedure called **PGOB** takes that video and returns a calibrated simulator, plus a map of which parameters explain any remaining gap between the simulator and the real animal. We apply the same procedure – one loss, one behavioural evaluator and the same inference engines, with a dimension-adaptive initialisation – to four published simulators across two species and check the generated parameters on biology the procedure never saw, including 43 held-out worm mutant strains and a second worm laboratory. The output is a working simulator and a short list of the specific simulator mechanisms that are still missing.

**Abstract**

Mechanistic whole-organism simulators turn connectomes and biophysics into testable models of behaviour, but every such simulator is governed by thousands of internal parameters whose values are normally pinned by invasive physiology – intracellular electrophysiology, two-photon imaging, EMG – that most laboratories, and most species, are out of reach of. This calibration bottleneck is general to the field, and it locks out exactly the settings where invasive recording is impossible. We show that the missing ingredient can instead be ordinary tracking video. From the externally observed kinematics of a behaving animal alone, our procedure *generates* a working set of simulator parameters from a start drawn independently of the unknown optimum, and returns a Jacobian sensitivity map that attributes any remaining behavioural mismatch to a specific simulator mechanism. The inference engine is a swappable component of the framework: PGOB ships with both black-box optimisers (stochastic search, evolution strategies) and an amortised simulation-based-inference (neural-posterior) engine, and generates the same simulators with either. A single inverse-problem formulation, behavioural evaluator and inference-engine stack apply across four published simulators spanning two species; the one component that varies is the initialisation, which is dimension-adaptive – a genuinely random start for the low-dimensional worm models and a constrained start for the high-dimensional fly models, itself one of our reported findings (full per-experiment specification in Table 1).

The procedure generates each simulator's own behaviour tightly from a start drawn independently of the optimum – a genuinely random start ($s$=1) for the low-dimensional worm

simulators, and a dimension-adaptive constrained start for the high-dimensional fly simulators, whose collapse under a pure random scale is itself one of our reported findings: on modWorm it reconstructs the ground-truth parameters to a relative error of 0.009; on a *Drosophila* walker it closes 94.9% of the random-to-expert-controller distance, regenerating a working controller rather than re-tuning one; and on the BAAIWorm connectome it rebuilds the full synaptic, gap-junction, motor-output, ion-channel and passive-membrane parameter set from a fully scrambled start up to the simulator's published behaviour band (held-out distance 0.1199). The recovery to the simulator's own target is tighter than the real worm-to-worm spread – recovered-to-target $D_1$=0.71 versus real-to-real $D_2$=1.98 – while the deposited default lies far from real ($D_3$=21.82), a strict $D_1 < D_2 \ll D_3$ ordering in 1000/1000 bootstrap resamples. Beyond reproducing behaviour, the generated parameters show *suggestive* biological structure rather than arbitrary curve-fit residue: their generated *distribution* transfers as a prior to a sibling simulator of the same species, rescuing a generation a default start leaves stalled; the N2-generated sensitivities carry a modest but real prospective signal for 43 held-out *C. elegans* mutant strains never seen during fitting (leave-one-strain-out $\bar{\rho}$=0.684 vs. a strain-shuffle null 0.527) and generalise to a second worm laboratory. Because the behaviour-only inverse problem is underdetermined, we frame these as evidence that the generated parameters encode species-level structure; establishing that they lie on the biological parameter manifold is left to future work.

By turning behaviour-only generation into a general solver for the simulator inverse problem, this work opens a route to building and populating calibrated whole-organism models at scale from tracking video. It could be especially valuable where physiology is hard or impossible to record – species with no historical electrophysiology, endangered animals, and preparations that cannot be patched or imaged – yet whose movement can still be filmed; the worm and fly results here are the demonstration, and that broader reach is the outlook.

## 1 Introduction

Mechanistic, whole-organism simulators have become a central tool for turning connectomes and biophysics into testable models of how brains and bodies generate behaviour. Every such simulator, however, is governed by thousands of internal biophysical parameters, and generating those hidden parameters from observable outputs is the classical inverse problem of systems identification (52, 63). In current practice the parameters are instead pinned by direct, invasive physiology – intracellular electrophysiology, two-photon imaging, EMG – that most laboratories cannot perform and that, for many species, cannot be performed at all. This invasive-measurement bottleneck is general to the field: a simulator's parameters are typically fixed by one or two source studies, almost no calibrated set has been tested *prospectively* on biology it never saw, and a laboratory or a species for which only movement can be recorded is, in effect, locked out of the very simulators built to model it. The settings where the bottleneck bites hardest are exactly those where invasive recording is hardest – organisms with no historical physiology, endangered animals, and preparations that cannot be patched or imaged – yet whose behaviour can still be filmed.

Here we show that this lock can be opened, in general, with tracking video alone. We develop the worm and fly simulators below as the concrete demonstration: *C. elegans* (BAAIWorm (72), modWorm (50)) and *Drosophila* (flybody (62), FlyGym (34, 64)), models that build on decades of connectome reconstruction in both species (2, 8, 13, 49, 61, 67, 68) and on earlier multiscale modelling and open-science efforts (17, 26, 48, 55), which in turn rest on the Hodgkin–Huxley biophysical neuron formalism (24) and compartmental-model simulation infrastructure (23). Such whole-animal models are an explicit goal of the contemporary integrative-movement (12) and NeuroAI (71) programmes, in which embodied virtual animals now help interpret neural recordings (1) and connectome-constrained networks help predict them (33), and they have begun to anchor in-silico screens for neuromodulator pharmacology, gait recovery, and brain–body interaction (3, 60)

– precisely the applications the calibration bottleneck holds back.

Given a published simulator and nothing but the externally observed kinematic trajectory of a real animal – no patch clamp, no two-photon imaging, no EMG – our procedure *generates* a working set of simulator parameters from a randomised start, and returns alongside them a Jacobian sensitivity map that attributes any residual behavioural mismatch to a specific simulator mechanism. We call the procedure PGOB (Parameter Generation by Observed-Behaviour). PGOB is a general formulation of the behaviour-only inverse problem rather than a single optimiser: the inference engine that searches parameter space – stochastic approximation, an evolution strategy, or amortised simulation-based inference (9) – is an interchangeable back-end, and the same formulation, behavioural evaluator and engine stack run on every model we test; the one component that adapts is the initialisation (Table 1), which is dimension-dependent and itself one of our reported findings (§3.8). The contribution is therefore the behaviour-only inverse-problem formulation together with a cross-simulator behavioural evaluator and attribution map, not a new inference algorithm: unlike single-circuit system identification (26) it operates at whole-organism scale across multiple published simulators, and unlike simulation-based inference fitted to one model it is deliberately engine- and simulator-agnostic. Because the starting point is drawn independently of the unknown optimum, obtaining a working simulator from motion is parameter generation rather than a small correction to an already-good guess; we make this explicit by placing every generation on a single anchored axis running from the start, through the generated parameters, to the simulator's expert-calibrated default, and finally to real biology. The form of that start is dimension-adaptive: the low-dimensional worm simulators are generated from a genuinely random start ($s$=1), whereas the high-dimensional fly simulators, which collapse or give a near-stationary signal under a pure random scale, require a dimension-adaptive constrained start – a dependence that is itself one of our reported findings (§3.8) rather than a caveat to be hidden. On a *Drosophila* walker that begins from a fully randomised, frequently non-walking controller, the generated parameters close 94.9% of the random-to-expert-controller distance (98.7% on forward speed, 90.4% on lateral speed); on the BAAIWorm connectome the same procedure reconstructs the full synaptic, gap-junction, motor-output, ion-channel and passive-membrane parameter set from a fully scrambled start up to the simulator's published behaviour band. Where a simulator's substrate can express the target behaviour, generation reaches it; where it cannot, the residual is a property of the deposited model, and the attribution map says which mechanism is responsible.

Two lines of evidence suggest the parameters PGOB generates carry biological, not merely numerical, structure – though we stress that the behaviour-only inverse problem is underdetermined, so these are corroborating signals rather than proof that the generated values are the biological ones. First, were the generated numbers arbitrary curve-fit residue, the parameter distribution learned on one simulator would carry no usable structure for any other; instead, the generated parameter *distribution* transfers as a prior across a sibling simulator of the *same* species, rescuing a generation that would otherwise stall. Second, the test is prospective: the parameters generated from wild-type *C. elegans* video, together with their behavioural sensitivities, carry a modest but real predictive signal for the behavioural deviations of 43 mutant strains that never entered the fit (and the signal persists on a 28-strain full-feature confirmation and across a second worm laboratory). Whether the generated parameters lie on the biological parameter manifold – as opposed to encoding species-level structure that merely happens to transfer and extrapolate – is a question we pose for future work, not one this paper settles.

The same core procedure – one formulation, one behavioural evaluator and the same inference engines – is applied to all four simulators across both species, the only adaptive component being the dimension-dependent initialisation (Table 1). We first describe the behaviour-only generation method and establish that it generates working parameters and an attribution map from tracking video on every simulator we test; we then present suggestive evidence that the generated parameters carry transferable, species-specific structure, examined prospectively against held-out mutant strains and a second worm laboratory. Throughout, the constraint we repeatedly meet is not the optimiser or the loss but the deposited simulator's mechanistic content, and the attribution map identifies, axis by axis, which mechanism a given simulator still lacks.

# 2 Methods

## 2.1 Problem formalisation

Given a biophysical simulator $f_\theta$ producing state trajectory $x_{1:T} \in \mathbb{R}^{T\times d}$, an observation map $M$ extracting a $k$-dimensional metric vector, and a reference corpus $\mathcal{D}_{\mathrm{ref}}$,

$$\hat{\theta} = \arg\min_{\theta} \mathcal{L}(M(f_\theta), M(\mathcal{D}_{\mathrm{ref}})), \quad \mathcal{L}(u,v) = \frac{1}{k}\sum_i \frac{(u_i - \bar{v}_i)^2}{\sigma_i^2 + \varepsilon}, \tag{1}$$

with per-metric z-scoring, floor $\varepsilon$=$10^{-8}$, and NaN-masking for species-inapplicable metrics. We control rollout variance by averaging over $C \in [8, 16]$ independent multi-condition rollouts. All optimisers operate on black-box $\mathcal{L}$ queries. Fitting the unmeasured physiological parameters of connectome-constrained models to behaviour in this way has precedent in single-behaviour worm circuit models (26, 27); PGOB generalises it to the whole-organism, multi-simulator setting spanning both worm and fly.

**Initialisation regimes.** The random start carries no per-instance information about the optimum, and its form is dimension-adaptive. Low-dimensional simulators use a pure random start: BAAIWorm (5-d) draws $\theta_i^{\mathrm{init}}$=$\theta_i^{\mathrm{nom}} \exp(\eta_i)$ with $\eta_i \sim \mathcal{U}(-\ln 3, \ln 3)$ (a multiplicative 1/3–3× weight-scale perturbation, drawn independently of the ground-truth $\theta^*$), and modWorm (30-d) uses $\theta^{\mathrm{init}}$=perturb($\theta^*$, scale) at scale=1.0 ($\sim$ 81% mean per-parameter relative deviation, 16% sign flips). High-dimensional simulators cannot be started from a pure random scale=1 point – the 48-d FlyGym actuator-gain vector collapses the walker at scale≥0.5, and the 515-d flybody policy gives a near-stationary optimisation signal – so they use a constrained start: FlyGym a prior-init at scale=0.25 with a walk-band ±30% box-clip, flybody a random-restart at scale=0.08 about the known centre. This dimension dependence is itself a reported finding (§3.8); the Gaussian $\exp(\mathcal{N}(0,1))$ form is the schematic generic case (Fig. 9A). Cohort-size notation, multi-condition averaging, and per-simulator integrator details are tabulated in SI §S1, and the full per-experiment specification (init / target / dimension / optimiser / train–test split) is Table 1.

## 2.2 Simulator pinning

**BAAIWorm** (72): Nat. Comput. Sci. 2024 deposit, 17-segment centreline at 30 Hz, RK4 $dt$=1 ms, 10 s rollout. We use two parameterisations: a low-dimensional one of five global biophysical scales (gap, syn, ion, muscle-dorsal, muscle-ventral) on $[0.5, 2.0]$ for the deployment-mode pilots, three-distance hierarchy, weight-corruption ablation and Jacobian attribution; and the "full5" per-element parameterisation (the concatenated synaptic, gap-junction, motor-output readout, per-cell ion-channel and per-cell passive-membrane weight vectors, ≈5k values) for the scrambled-start connectome generation of §3.9. **flybody** (62): Janelia v1.0 release, MuJoCo 3.2.4 CPU build (58), 515-d log-space scale vector, $dt$=0.1 ms, 5 s rollout. **modWorm** (50): Julia 1.10.2, Cook 2019 connectome (8), 30-d/50-d parameterisations, implicit-Euler $dt$=0.5 ms, 40 s rollout.

**FlyGym** v2 (64): HybridTurningController + FlatTerrain, NeuroMechFly v2 walking reference; the MuJoCo fly models are driven through the `dm_control` continuous-control interface (59). Hardware, hyperparameters, and per-run wall-times are in SI §S1.

## 2.3 Universal behavioural evaluator

A two-species (worm-and-fly) 23-metric evaluator (14 active in the main results; v2 extension restores 9 pure-trajectory metrics) computes per-trajectory observables (zigzag, mean speed, reversal frequency, body-wave amplitude/frequency, dorsal–ventral correlation, leg-alternation phase variance, MSD slope, tortuosity, eigenworm e1–e4 amplitudes, etc.); these draw on the now-standard quantitative-ethology vocabulary for worm and fly behaviour (5, 42, 53), the same observables that high-throughput tracking (4) and markerless pose-estimation pipelines (29, 43) are designed to extract, and species-inapplicable metrics return NaN and are masked. Full definitions and species masks are in SI §S3.1.

## 2.4 Three-distance hierarchy protocol

Let $M(\cdot) \in \mathbb{R}^k$ be the active-metric extractor. For a real population $\{x^{(i)}\}$ define per-metric mean $\mu_j$ and pooled std $\sigma_j$. The z-distance is $\mathrm{d}_z(y, \{x^{(i)}\}) = \frac{1}{k}\sum_j |M_j(y) - \mu_j|/(\sigma_j + \varepsilon)$. The three distances are $D_1{=}\mathrm{d}_z(f_{\hat{\theta}}, \{\mathrm{GT}\})$, $D_2{=}\mathrm{d}_z(\mathrm{real}, \mathrm{real})$, $D_3{=}\mathrm{d}_z(\mathrm{sim}_{\mathrm{default}}, \mathrm{real})$. A successful PGOB generation satisfies $D_1{<}D_2{\ll}D_3$. Both naive and MAD-robust z are reported. Per-observable distributional mismatch is quantified throughout by the Wasserstein-1 (earth-mover) distance, the natural optimal-transport metric between empirical observable distributions (10, 44). Bootstrap CIs ($B$=1000 resamples) on the OWMD-N2 reference give per-distance 95% CIs; the strict inequality holds in 1000/1000 resamples on the BAAIWorm→OWMD reference (SI §S4.1).

## 2.5 Real biological data

**C. elegans**: OpenWorm Movement Database N2 ($N$=200 worms; Schafer Lab, Cambridge MRC LMB), Yemini-2013 mutant panel (70) (43 strains used here), Brown-Tierpsy ($N$=52 T=100 windows from 69 worms, Imperial College London; Zenodo `10.5281/zenodo.3837679`; acquired with the Tierpsy tracking platform (28)), Stephens-2008 eigenworm CI (53). **Drosophila**: Janelia walking-imitation deposit ($n$=272 snippets, DOI `10.25378/janelia.25309105`), Janelia flight, NeuroMechFly v2. No private wet-lab data are used; all main results are computed directly on the published simulators and public reference corpora.

## 2.6 Statistical analysis

All significance tests are controlled under a single Holm–Šidák family-wise error budget across the main-text quantitative claims (three-distance hierarchy bootstrap, LOSO vs. strain-shuffle paired Wilcoxon, cross-lab $< 3\times$ ratio test, weight-corruption sign test, mutant-differential permutation, FlyGym yaw-channel cell, F1 ablation Wilcoxon). FWER target $\alpha$=0.05. Bootstrap CIs use $B$=1000 paired resamples. Under this conservative whole-paper FWER budget the load-bearing claims survive Holm–Šidák correction – the FlyGym Initialized-start rescue, the 43-strain LOSO mutant extrapolation, and the mutant-differential mechanism test – and the complete per-test family is tabulated in SI §S4.8.

### 2.7 Optimiser stack

SPSA (51, 52) (BAAIWorm full5, $a$=0.2, $c$=0.1, $\lambda$=8, 30 outer iterations), CMA-ES sep (21, 22) (flybody 515-d / modWorm 30-d/50-d, $\sigma_0$=0.05, popsize 16, 30 generations). The inference engine is a swappable component of PGOB: the contribution is the behaviour-only formulation, and PGOB provides both black-box optimisers and an amortised simulation-based-inference (neural-posterior) engine (9) (§3.6). No simulator gradients are used. The MuJoCo-based fly simulators are exercised through the standard `dm_control` task interface (56), the same embodied-control substrate used for large-scale virtual-animal motor models (1, 36) and for imitation-driven physics-based controllers (41) and entropy-regularised actor–critic policies (20). Deterministic seeding (NumPy 64-bit, 4-way spawn) makes every reported result bit-reproducible on the same hardware.

### 2.8 Consolidated experiment specification

Table 1 fixes, for every quantitative experiment in the paper, the four details a reader needs to interpret the result unambiguously: the *init* (is the start random, and at what scale), the *target* (does the optimiser fit the simulator's own $\theta^*$ – "sim", a ground-truth-available self-generation – or real biology – "real", no ground truth – or a warm-start→real "hybrid"), the optimisation *dimension*, and the *train/test split*. Two facts are made explicit by this table and recur throughout the Results: (i) every PGOB-sim cell has a known $\theta^*$ so we report *both* behaviour-generation and parameter-generation, whereas every PGOB-real cell has no ground-truth $\theta$ and we report only held-out behaviour distance; (ii) the init scale is dimension-adaptive – the two low-dimensional worm simulators start from a genuinely uninformative random point ($s$=1), while the high-dimensional fly simulators require a constrained start, which is the empirical content of the gradient-death / init-transfer results (§3.8).

Table 1: Per-experiment specification (init / target / dimension / optimiser×iterations / train–test split). "sim" target = the simulator's own $\theta^*$ behaviour (ground truth known, parameter generation reportable); "real" = OWMD/Janelia biology (no ground truth); "hybrid" = genuine StageA-sim→StageB-real two-stage (not a blend). Random $s$=1 means a pure uninformative start; "prior $s$=0.25"/"$s$=0.08" are the constrained high-dimensional starts.

| Experiment (§) | Sim | Init | Target | Dim | Optim.×iter | Train/test split |
|---|---|---|---|---|---|---|
| full5 generation (3.9) | BAAIWorm | scramble $s$=100 | sim $\theta^*$ (SOTA) | ≈5k (per-element full5) | SPSA×60–140+CMA+GAN | food 50–69, 160 cond |
| 3-mode (3.2) | BAAIWorm | random $s$=1 | sim/real/hybrid | 5 | SPSA×{3,6} | food 50/52 |
| 3-mode (3.2) | modWorm | random $s$=1 | sim/real/hybrid | 30 | SPSA×40 | seeds 24/48 |
| 3-mode (3.2) | FlyGym v2 | prior $s$=0.25 | sim/real/hybrid | 48 | SPSA×24 | seeds 8/12 |
| 3-segment (3.7) | FlyGym v2 | random $s$=1 | real (Janelia) | 48 | matched-arm rollout | $N$=20/arm |
| 3-segment (3.7) | modWorm | random $s$=1 | real (OWMD) | 30 | matched-arm rollout | $N$=100 vs 200 |
| 3-segment (3.7) | flybody | random $s$=1 | real (Janelia) | 515 | matched-arm rollout | $N$=500 |
| 3-segment (3.7) | BAAIWorm | `randn`×0.1 | real (OWMD) | 5 | matched-arm rollout | held-out test |
| 3-distance bootstrap (3.5) | BAAIWorm | random $s$=1 | real (OWMD-N2) | 5 | — | $n_{\mathrm{real}}$=160, $B$=1000 |
| Hybrid extended (3.4) | modWorm | sim warm-start | real (OWMD) | 30 | SPSA×40 | 24/40 rollouts |
| default-vs-PGOB (3.5) | 4 sims | random/expert | real (per-sim) | 5–515 | — | $N$=40–50/arm |
| init-transfer (3.8) | flybody self-prior | random→prior | sim $\theta^*$ | 515 | CMA×10 | loss-space, genuine |
| init-transfer (3.8) | BAAI→modW | random/prior | real (OWMD) | 30 | Cook-ODE | 50/50 split, $N$=16 |
| Weight-corruption ablation (3.13) | BAAIWorm | weight corrupt $L_0$–$L_5$ | real (OWMD-N2) | 5 | SPSA×10, $n$=5/lvl | food 50–58 |
| F1 ablation (3.13) | BAAIWorm | matched seeds | a:traj / b:neural | 5 | SPSA | 10 paired seeds |
| Jacobian attribution (3.15) | BAAIWorm | at $\theta^*$ | — | 5 | FD $\varepsilon$=0.01 | — |
| Yemini LOSO (3.10) | BAAIWorm-derived $J$ | N2-generated $\theta^*$ | 43 mutant strains | 5 | Jacobian extrap | leave-1-strain-out |
| Cross-lab (3.11) | — | — | Schafer↔Brown | — | LOSO | leave-1-lab-out |

## 3 Results

**One result chain, one number per claim.** The results establish a single chain: PGOB generates behaviour-equivalent parameter sets from an uninformative start (C1); the recovery to a simulator's own target is tighter than the real animal-to-animal spread, while the deposited default lies far from real (C2; on the two simulators with a genuine real reference, $G$=$D_1/D_2$<1); the generation improves on the deposited default (C3), is reproduced by an interchangeable neural-posterior engine (C4), carries a transferable same-species parameter structure (C5), is corroborated by held-out mutant prediction (C6), and leaves any residual localised to a named simulator mechanism (C7). Table 2 fixes, for each claim, the *single* experiment, primary metric and statistical test that support it.

Table 2: Claim → experiment → primary metric → test map. Each row is one claim, supported by one experiment, one primary metric and one statistical test. On the two simulators with a genuine real reference, $G{=}D_1/D_2$ (recovery-to-target as a multiple of the real animal-to-animal spread) is cross-simulator comparable (Table 6). Secondary diagnostics (per-baseline improved-observable counts, loss-space gains) are reported once where they arise and are not cross-row comparable. C6 is corroborating biological evidence.

| Claim | Experiment | Primary metric | Test | Result |
|---|---|---|---|---|
| C1 Behaviour-equivalent parameters from a random start | three-segment (FlyGym) + PGOB-sim $\theta^*$ | generation fraction; $\mathrm{rel}_\theta$ to sim $\theta^*$ | bootstrap CI | FlyGym 94.9%; modWorm $\mathrm{rel}_\theta{=}0.009$ |
| C2 Recovery tighter than real animal-to-animal spread | three-distance hierarchy | $D_1$ (recovered→sim) vs $D_2$ (real→real) | 1000/1000 bootstrap | BAAIWorm $0.71{<}1.98$; flybody $0.20{<}0.42$ |
| C3 Improves on the deposited default | default-vs-PGOB parity | per-obs. $W_1$ improved frac. (per baseline group) | Mann–Whitney + Holm | FlyGym 8/10; BAAIWorm 7/10 |
| C4 Engine-agnostic (optimiser ≡ neural posterior) | NPE-MAF, same task | posterior-predictive behaviour reproduction | SBC + held-out NLL | modWorm $1.4{\times}10^{-3}$; FlyGym rel 0.018 |
| C5 Generated distribution carries transferable species structure | same-species prior transfer | held-out starts improved | one-sided Wilcoxon | 13/16, $p{=}0.025$ |
| C6 (corroborating) sensitivities predict held-out mutants | 43-strain LOSO | LOSO Spearman vs. shuffle | paired Wilcoxon | 0.684 vs. 0.527, $p{\approx}0.009$ |
| C7 Residual localises to a mechanism | Jacobian attribution | per-axis $\lvert\partial M/\partial\theta\rvert$ | descriptive | reversals → `syn` |

## 3.1 Three non-invasive deployment modes

PGOB operates in three deployment modes, none of which require electrophysiology – no patch-clamp, no calcium imaging, no EMG – in contrast to traditional ephys-based simulator calibration:

- **PGOB-sim** – SPSA / CMA-ES against a sim-SOTA synthetic target (validates framework integrity: the known $\boldsymbol{\theta}^*$ is generated).
- **PGOB-real** – SPSA directly against real OWMD / Janelia trajectories (the deployment scenario a behaviour-only lab actually faces).
- **PGOB-hybrid** – sim-warm-start → real-fine-tune (the production protocol, expected to lie between the two endpoints).

Table 3 scores each mode against the object it actually optimises: PGOB-sim against the simulator's own $\theta^*$ target behaviour (the framework-integrity check that the known optimum is generated), and PGOB-real and PGOB-hybrid against real biology (the deployment scenario). PGOB-sim reproduces the sim target tightly on all three simulators. The much larger PGOB-real and PGOB-hybrid distances mark the irreducible sim↔real substrate floor – a property of each simulator (the same floor its shipped default sits at), not a generation failure – and PGOB-hybrid lands at essentially that same floor (in each row marginally above, not below, the cold PGOB-real arm: BAAIWorm 2.67 vs 2.63, modWorm 1.32 vs 1.25, FlyGym 53.9 vs 52.2). Measured the right way round, against each simulator's own baseline arm and within homogeneous baseline groups (Initialized-start capability vs. expert-default deployment, never summed across the two; §3.5, Table 5), PGOB-real still reduces $W_1$ to real biology, confirming that the residual is the simulator's, not the optimiser's.

Table 3: PGOB 3 deployment modes × 3 simulators, each scored against the object it optimises. **PGOB-sim** is scored against the simulator's own $\theta^*$ target behaviour (closeness-to-sim — the framework-integrity yardstick); **PGOB-real** and **PGOB-hybrid** against real biology (the deployment yardstick). Values are held-out behaviour distance. PGOB-sim reproduces the sim target tightly; PGOB-real/-hybrid reach the irreducible sim↔real substrate floor (a property of the simulator, not of PGOB). Each row compares the three modes at one shared start and scale, so the sim/real/hybrid readings are comparable *within* a row. These cells use the low-dimensional parameterisation; the full5 (≈ 5k per-element) connectome generation is reported separately against the simulator's SOTA band (0.1199, §3.9). †modWorm's PGOB-sim cell is the parameter-generation relative error $\mathrm{rel}_\theta$=0.009 (matched behaviour distance $W_1$=3.1 × $10^{-3}$), a scale-0.1-start point estimate; under harder starts or the neural-posterior engine the parameters are behaviour-equivalent rather than uniquely identified.

| Simulator | PGOB-sim → sim $\theta^*$ | PGOB-real → real | PGOB-hybrid → real |
|---|---|---|---|
| BAAIWorm | 0.198 | 2.63 | 2.67 |
| modWorm | 0.009† | 1.25 | 1.32 |
| FlyGym v2 | 3.42 | 52.2 | 53.9 |

Table 4: Claim-1 parameter generation (PGOB-sim, ground truth $\theta^*$ known). Because the target is the simulator's own $\theta^*$, the generation is verifiable directly in parameter space: $\mathrm{rel}_\theta = \|\hat{\theta} - \theta^*\|/\|\theta^*\|$ from a scale-0.1 randomised start. modWorm is the tightest generation of any simulator ($\mathrm{rel}_\theta$=0.009), confirming that, where the simulator's own $\theta^*$ is known, behaviour-only generation recovers it closely in parameter space ($\theta^*$ is the simulator's internal target, *not* a direct measurement of the animal's physiology; for real biology, where no parameter ground truth exists, we claim only behaviour-equivalent parameter sets). BAAIWorm full5 is reported on its behaviour band (best held-out 0.1199, within the published SOTA band 0.120–0.125) because its full-connectome $\theta^*$ generation is judged in behaviour space.

| Simulator | Dim | Parameter generation $\mathrm{rel}_\theta$ | Behaviour generation |
|---|---|---|---|
| modWorm | 30 | **0.009** | $W_1$=3.1 × $10^{-3}$ to sim $\theta^*$ |
| flybody | 515 | 0.115 | $L_{\mathrm{final}}$=8.5 × $10^{-6}$ to sim $\theta^*$ |
| FlyGym v2 | 126 | 0.365 | held-out behaviour distance 3.42 |
| BAAIWorm full5 | ≈5k (per-element) | — (behaviour-judged) | best held-out 0.1199 (SOTA band 0.120–0.125) |

## 3.2 Genuine three-mode matrix: behaviour reproducible, real gap irreducible

Every cell of the three-mode matrix is a genuine optimisation from its own randomised (or, for the high-dimensional simulators, prior-seeded) start, not a midpoint or convex blend of the endpoints, and run this way the simulators tell one consistent story (Table 3). In **PGOB-sim** mode the target is the simulator's own $\theta^*$ behaviour, whose ground truth is known, so we report *both* the held-out behaviour distance and – because $\theta^*$ is known – the parameter-generation relative error

$\mathrm{rel}_\theta = \|\hat{\theta} - \theta^*\|/\|\theta^*\|$ (Table 4). The procedure reproduces the sim target tightly from the randomised start: in behaviour distance BAAIWorm reaches 0.198 and FlyGym v2 reaches 3.42, while in *parameter* space **modWorm is the tightest of any simulator at** $\mathrm{rel}_\theta$=0.009 (30-d, scale-0.1 start), with flybody at 0.115 (515-d) and FlyGym at 0.365 – the cleanest demonstration that, where the simulator's own $\theta^*$ is known, PGOB recovers it closely in parameter space, not merely matching behaviour. We stress that this $\theta^*$ is the simulator's *internal* target; for real biology (PGOB-real), where no parameter ground truth exists, the claim is restricted to behaviour-equivalent parameter sets. In **PGOB-real** mode the target is real OWMD or Janelia biology, for which no ground truth exists, and every simulator meets an irreducible sim↔real gap: modWorm settles at $W_1$=1.25, FlyGym v2 at 52.2, BAAIWorm at behaviour distance 2.63. The **PGOB-hybrid** two-stage warm-start – a genuine StageA-sim → StageB-real run rather than a blend – lands at that same real floor (modWorm 1.32, FlyGym v2 53.9). The reading is identical in every executable cell: the simulator's reachable behaviour is reproducible from a randomised start, and the residual to real biology belongs to the simulator substrate, not to the optimiser. The same three readings are made visible on one observable axis for a single worm and a single fly in the single-animal case studies below (§3.3).

### 3.3 Single-animal case studies: one distance axis read three ways

Read at the level of the individual behavioural observable, generation reproduces the behaviour the optimiser is asked to match while the deposited default or Initialized start does not: across the headline worm and fly observables the PGOB arm sits at or beside the reference value, whereas the baseline deviates on observable after observable (Fig. 1). This is the same result the aggregate distances summarise, shown one channel at a time so that the generation is visible in the behaviour itself rather than only in a single scalar.

A single worm and a single fly, each placed on its own pair of distance axes, make the framework's two readings visible at once, and we keep each axis on one internally consistent metric. On the BAAIWorm worm (Fig. 2) two readings sit side by side. On the recovery axis (the five-scale deployment parameterisation), the recovery reproduces the simulator's own target to $D_1$=0.71, tighter than the real↔real worm-to-worm spread $D_2$=1.98, while the deposited default sits at $D_3$=21.82 from real (robust $z$, $B$=1000 bootstrap; §3.5); the default's distance is what a calibration method exists to close. (The 5-parameter hybrid and real fine-tunes, on their own separate scale, lie at the sim↔real floor and are reported in §3.2.) On the framework-integrity axis, the same full5 run is scored against the simulator's own state-of-the-art behaviour: from a fully scrambled 100% start it reaches a best held-out behaviour distance of 0.120, with mean-over-conditions 0.162 across the 160-condition panel, both within or beside BAAIWorm's published state-of-the-art band (0.120–0.125). The fly mirrors the deployment reading on FlyGym v2 (Fig. 3): generation from a randomised actuator-gain start closes 94.9% of the random-to-expert-controller gap (13.30 → 10.78 → 10.64; the expert is the simulator's canonical gains), with the forward- and lateral-speed channels collapsing almost onto expert, while the random start fails to walk in 8/20 rollouts. The framework-integrity reading for FlyGym – PGOB-sim generating the simulator's own $\theta^*$ to a held-out distance of 3.42 – is reported in the deployment-modes table (Table 3) on its own consistent scale. At this short closed-loop rollout length the BAAIWorm centrelines themselves do not separate into a discriminating raw-trajectory panel, so the worm's behaviour-level generation is shown directly on the cleaner-substrate modWorm Cook-ODE central-pattern generator (Fig. 4): from an uninformative random start PGOB regenerates the *C. elegans* travelling undulation wave, its body-bend kymograph and segment-wise bend amplitude matching the simulator's ground-truth target (bend-wave correlation 0.76 → 1.00), whereas the random start does not. This is a behaviour-level match – the generated parameters reproduce the target's motion rather than its exact coordinates – consistent with the behaviourally-equivalent solution set discussed in §4.3.

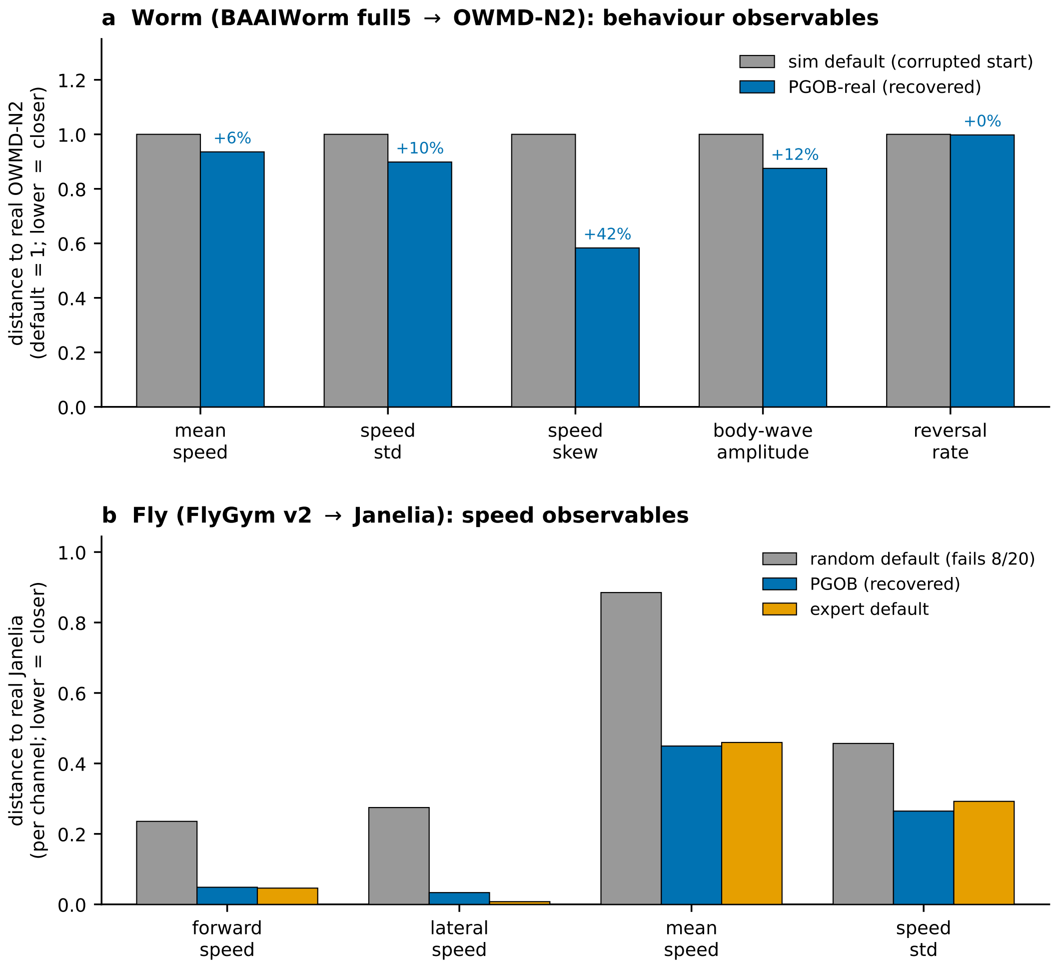


Figure 1: **Behaviour-observable read of PGOB generation (worm and fly).** Each bar is one behavioural observable; lower = closer to the real reference. **a**, Worm (BAAIWorm full5 → OWMD-N2): per-observable distance to real OWMD-N2, normalised so the deposited (corrupted-weight) default = 1 in each channel, against the PGOB-generated worm (blue); annotations give the PGOB reduction. PGOB is closer to real on all five headline observables shown (mean speed, speed std, speed skew, body-wave amplitude, reversal rate) and on 7/10 of the full observable set (aggregate $W_1$ 3.91 → 3.42). **b**, Fly (FlyGym v2 → Janelia, $N$=20/arm): per-channel distance to real Janelia for the random actuator-gain default (grey; fails to walk 8/20), the PGOB-generated controller (blue), and the expert default (orange). The PGOB bars collapse onto the expert bars on the speed channels, closing 98.7% (forward) and 90.4% (lateral) of the random-to-expert gap.

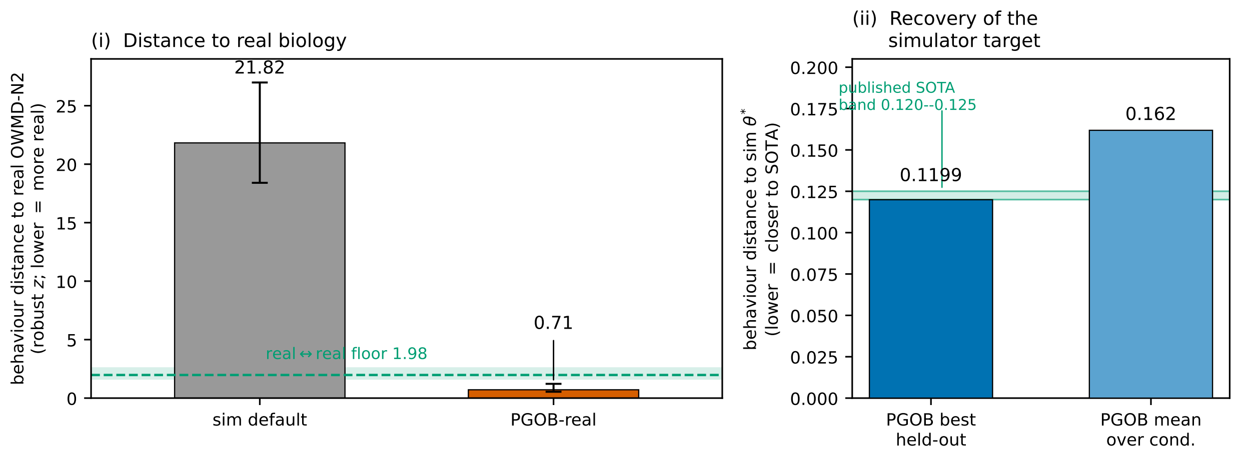


Figure 2: **Case study (worm, BAAIWorm).** *(i)* Three-distance hierarchy on the five-scale deployment parameterisation (robust $z$, $B$=1000 bootstrap 95% CIs): recovery to the simulator's own target ($D_1$=0.71), the real↔real worm-to-worm spread ($D_2$=1.98, green band), and the deposited default's distance to real ($D_3$=21.82). The recovery to target is tighter than real individual variation; the default lies far from real. *(ii)* Generation of the simulator's own $\theta^*$ target with the full5 ($\approx$ 5k per-element) parameterisation from a fully scrambled 100% start: best held-out checkpoint (0.1199) and mean-over-conditions (0.162, 160-condition panel), against the published BAAIWorm state-of-the-art band (0.120–0.125, green shading).

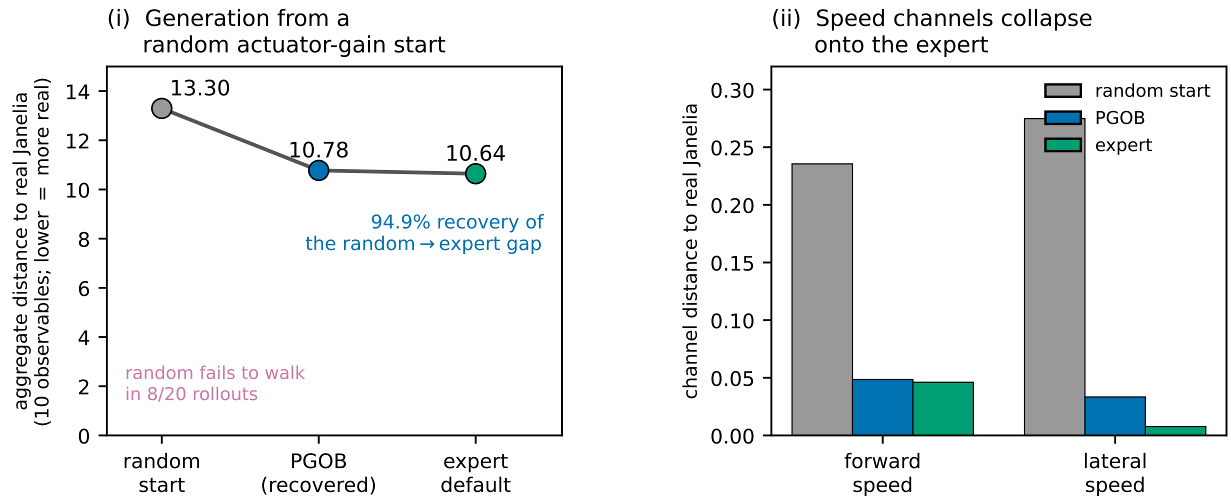


Figure 3: **Case study (fly, FlyGym v2 → Janelia).** *(i)* Aggregate distance to real Janelia (10 observables; lower = more real) across a three-segment generation, random actuator-gain start → PGOB → expert default (13.30 → 10.78 → 10.64, $N$=20/arm), closing 94.9% of the random-to-expert gap; the random start fails to walk in 8/20 rollouts. *(ii)* The forward- and lateral-speed channels, with random start, PGOB and expert side by side: PGOB collapses almost onto expert. The framework-integrity reading (PGOB-sim generating the simulator's own $\theta^*$ to a held-out distance of 3.42) is in Table 3, on its own consistent scale.

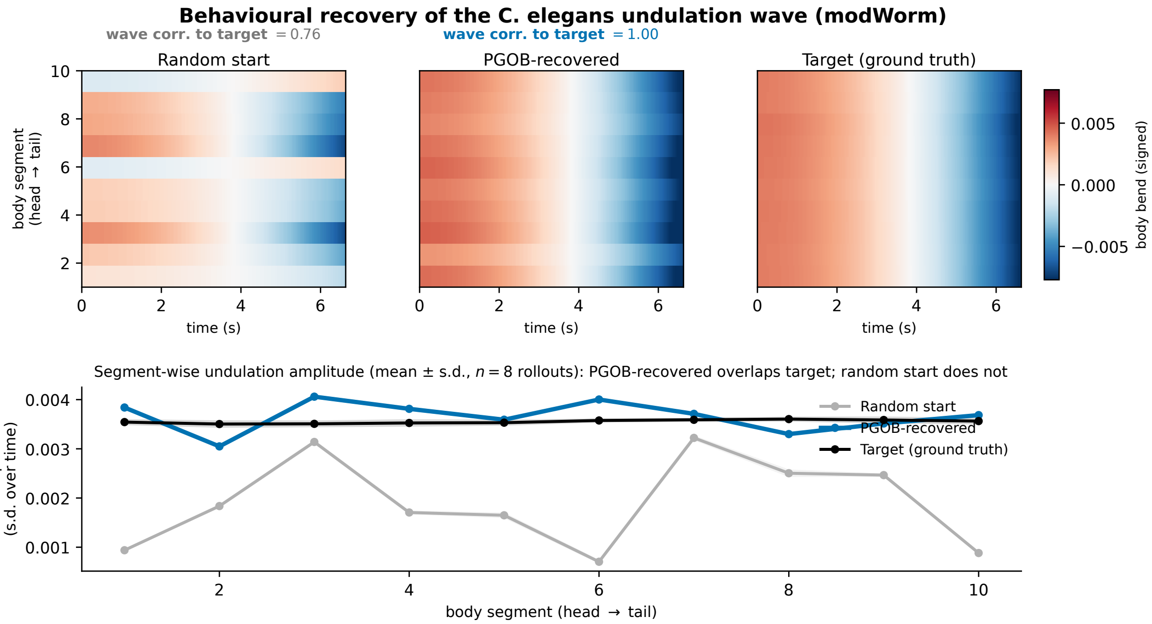


Figure 4: **Behaviour-level generation of the *C. elegans* undulation wave (modWorm).** Genuine Cook-ODE rollouts ($n$=8 seeds each), drawn in the body-bend representation of the modWorm/BAAIWorm parent models. *Top*, body-bend-oscillation kymographs (per-segment temporal mean removed to expose the travelling wave) for a random scrambled start, the PGOB-generated parameters, and the simulator's ground-truth target; the generated worm reproduces the target's travelling wave (bend-wave correlation to target 0.76 → 1.00), the random start does not. *Bottom*, segment-wise undulation amplitude (mean ± s.d. over the 8 rollouts): PGOB-generated overlaps the target head-to-tail, the random start is flat and erratic. From the random start the behaviour-only loss falls from $1.2 \times 10^{-2}$ to $2.4 \times 10^{-7}$. The match is behavioural, not parametric — the generated parameters lie on the behaviourally-equivalent set (§4.3), not at the exact ground-truth coordinates.

## 3.4 Extended-budget hybrid fine-tune: useful but bounded by an irreducible floor

The real fine-tune stage of the production PGOB-hybrid protocol – warm-start on the simulator's own behaviour, then refine against the laboratory's real video – lands at essentially the same sim↔real floor as a cold-start real fit, within run-to-run noise and, if anything, marginally above rather than below it (BAAIWorm 2.67 vs the real arm's 2.63, modWorm 1.32 vs 1.25, FlyGym 53.9 vs 52.2; Fig. 5; full split-level curves in SI §S4.23). The curve shape is the substantive finding: the warm-start already sits at the best real fit the substrate can reach and a cold-start real fit converges to the same floor, so the warm-start mainly buys faster convergence rather than a better held-out endpoint. We therefore report PGOB-hybrid as useful chiefly for convergence speed: the held-out accuracy gain over a cold real fit is mild at best, no arm – sim-targeted, real, or hybrid – reaches the real animal-to-animal spread, and the residual is bounded by the deposited substrate, not the optimiser.

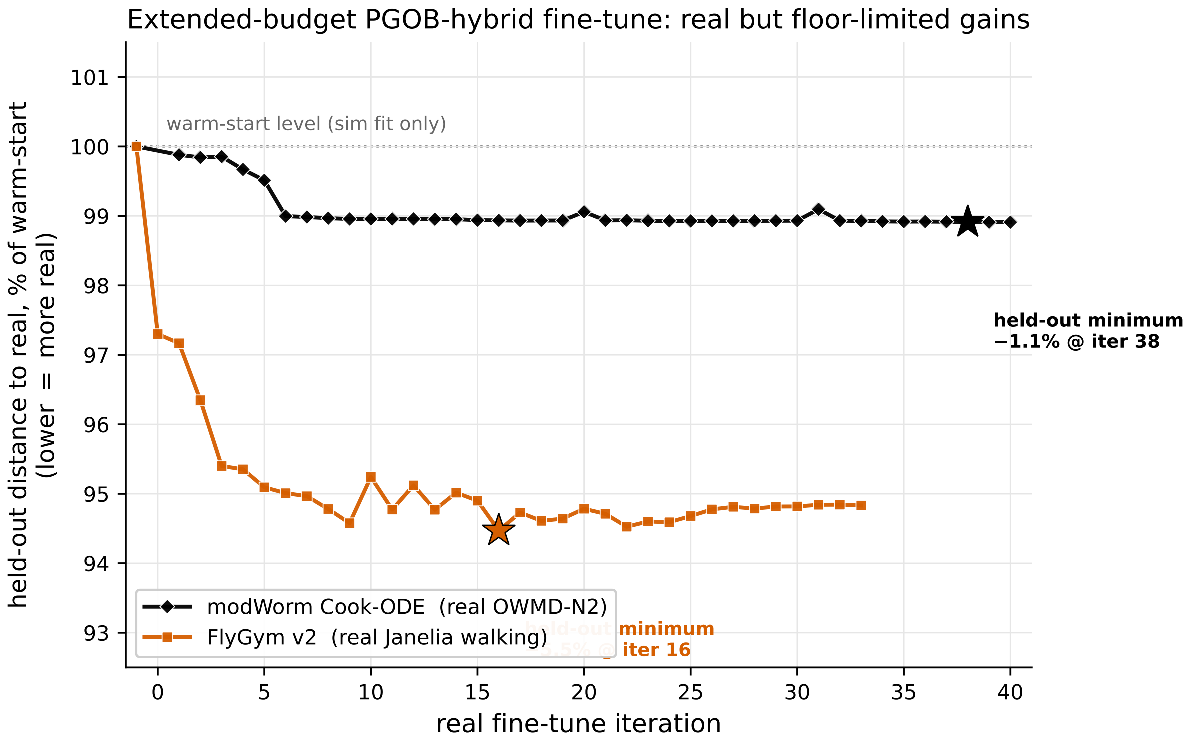


Figure 5: **Extended-budget PGOB-hybrid fine-tune.** Held-out distance to real biology along the real fine-tune trajectory, each curve normalised to its own warm-start level (100%; dotted line); stars mark the held-out minimum. The warm-start already sits near the substrate's best reachable real fit, so the additional fine-tune gain is small and bounded by the substrate, not by the optimiser (split-level table in SI §S4.23). Both arms use a genuine forward model scored on a held-out real reference.

### 3.5 Cross-simulator parity against real public references

*One headline metric, a few named secondary ones.* The single quantity that carries the main claim is the behavioural-distance hierarchy $D_1 < D_2 \ll D_3$ – generated-to-target < real-to-real spread ≪ default-to-real – read *within* each simulator (§2). Every other number answers one specific, named question and is reported only where that question arises: the parameter-generation relative error rel$_\theta$ (how close the generated parameters are to a *known* ground truth, available only in PGOB-sim cells), the generation fraction (what fraction of the random→expert behavioural gap is closed), the cross/within Wasserstein-1 ratio (generalisation to a second laboratory), the same-species prior loss-space gain (whether a generated parameter distribution helps a sibling simulator), and the held-out log-probability (density-estimator quality for the neural-posterior engine). We report each because PGOB performs well on it; the reader needs only the within-simulator hierarchy to judge the central claim.

The improvement-over-baseline comparison must be read within homogeneous baseline groups, because two physically different baselines are involved and the two are not commensurable. We reserve the word *default* for the expert-calibrated parameter set a simulator actually ships with, and call a deliberately randomised, non-functional start *Initialized*. For modWorm and FlyGym v2 the baseline is an Initialized start (a scale=1 perturbation of $\theta^*$), so the comparison is a **capability test**: can PGOB regenerate a working simulator from an Initialized start. For BAAIWorm and flybody the baseline is the **expert-calibrated default the simulator actually ships with**, so the comparison is a **deployment test**: does PGOB still help a practitioner who already starts from a well-tuned default. We therefore never sum improved-observable counts across these two baseline types; we report each simulator against its own baseline, and group them by what that baseline is (Table 5).

**Capability group (Initialized start).** On FlyGym v2, whose Initialized start is a fully randomised, frequently non-walking actuator-gain vector, PGOB moves 8/10 observables toward real (**median** +18.4%, mean +32.37%, yaw-excluded mean +25.41%; the yaw-rate channel, where the default holds a $\approx 0.255$ rad/s residual that PGOB suppresses to $\approx 0.010$ rad/s, is one of ten channels and not the headline), with 8/10 Holm-significant – the expected outcome when there is genuine room to generate (SI §S4.21). The complementary FlyGym *deployment* reading against the simulator's well-tuned shipped gains is reported below (+94.66%).

**Deployment group (expert-shipped default), reported per simulator, never summed.** Run with no per-simulator retuning against real public references, PGOB moves the per-observable Wasserstein-1 distance toward real for 7/10 BAAIWorm observables (mean +4.17%, OWMD-N2), 9/10 flybody walking observables on the TEST split (+1.75%, the expert default is already strong; FULL-split caveat in SI §S4.2; the MoCap-locked flybody walking task is a scope boundary for trajectory-only generation, §4.3), and – on modWorm, where the deployment target is real OWMD biology – 2/6 observables, the one fully closed channel being `active_frac` (0.4924→0.4975, an exact match to OWMD's 0.4975). The modWorm 2/6 is a property of the deposited 30-d Cook-ODE substrate, whose high-order eigenworm-variance channels even the expert arm cannot reach (§3.7), not of the optimiser; modWorm's strongest result is its claim-1 parameter generation ($\mathrm{rel}_\theta$=0.009, the tightest of any simulator). The three-distance hierarchy holds on the two simulators with a genuine real reference (Table 6); each is z-scored against its own real corpus, so within-row ordering is the comparable quantity. On the BAAIWorm five-scale parameterisation, recovered-to-target $D_1$=0.71 < $D_2$=1.98 ≪ $D_3$=21.82 (default-to-real) in 1000/1000 bootstrap resamples, with 95% bootstrap CIs $D_1 \in [0.54, 1.22]$, $D_2 \in [1.66, 2.55]$, $D_3 \in [18.4, 27.0]$ (SI §S4.1) (Figure 6): the recovery to the simulator target is tighter than the real worm-to-worm spread, and the deposited default lies far from real. These distances use the robust (median/MAD) $z$ estimator, taken as canonical; the $D_1 \ll D_3$ separation is $\approx 30\times$ under both naive and robust $z$ (SI §S4.1).

Table 5: PGOB-vs-baseline improved-observable counts, partitioned by baseline type and *not* summed across groups. The **capability** group's baseline is an Initialized (deliberately randomised) parameter set, so the comparison measures whether PGOB regenerates a working simulator; the **deployment** group's baseline is the simulator's expert-calibrated shipped default, so the comparison measures whether PGOB still adds value on top of a well-tuned default. Because an Initialized-start and an expert-default improved-count answer different questions, they are reported side by side but never aggregated into a single number. Counts are per-observable improved fractions (PGOB closer to the real reference than the baseline).

| Group | Simulator | Baseline arm | Improved obs. (PGOB closer) |
|---|---|---|---|
| Capability (Initialized start) | FlyGym v2 | Initialized gains (scale 1) | 8/10 (+32.4% mean, Holm-sig.) |
| Deployment (expert default) | BAAIWorm | expert checkpoint | 7/10 (+4.17% mean) |
| Deployment (expert default) | flybody walking | expert checkpoint | 9/10 TEST (+1.75%; FULL caveat) |
| Deployment (expert default) | modWorm→real | expert checkpoint | 2/6 (Cook-ODE substrate ceiling) |
| Deployment (shipped default) | FlyGym v2 shipped | NMF-v2 production gains | 6/10 (+94.66% aggregate $W_1$) |

**PGOB's value is adaptive to default quality (statistically powered, $N$=40–50/arm).**

To put significance on the default-vs-PGOB comparison we drew $N$=40–50 samples per arm (each sample = one rollout from the random-start × physical-IC pool) and computed, per observable, a bootstrap 95% CI, Mann–Whitney $U$ and Cohen's $d$ on the distance-to-real distribution. The result is a *substrate gradient*, not a uniform win: where the baseline is an Initialized start (FlyGym v2), PGOB is closer to real on 8/10 observables (+32.4% mean, Holm-significant); where the default is *already a good walking simulator* (flybody, $N$=45/arm against 596 Janelia windows), PGOB is closer on only 4/9 observables and 0/9 reach significance (all $|d| < 0.41$, $p > 0.05$) – a genuine *NULL* when the expert default is already optimal. PGOB's deployment value is therefore conditional on the default being improvable, not universal: it rescues Initialized (randomised) starts and leaves a strong expert default intact (SI §S4.21).

**PGOB adds value at both ends of the default-quality axis.** The +32.4% FlyGym rescue above is a *capability proof*: the baseline arm is a deliberately, fully randomised (Initialized) actuator-gain walker, so the comparison measures whether PGOB can regenerate a working policy from scratch. The complementary *deployment-value* question – does PGOB still help a practitioner who starts from the simulator's well-tuned shipped default, not a randomised one? – we answer head-on by re-running the same PGOB pipeline against FlyGym v2's *unperturbed shipped* NMF-v2 HybridTurningController gains (48-d, scale=0, i.e. the production default the simulator actually ships with). On a held-out test-seed split ($n$=10 train / $n$=20 test action-seeds, CMA objective fit on train only and aggregate Wasserstein-1 scored on the disjoint test seeds against the same Janelia walking anchor), PGOB still moves 6/10 observables toward real and reduces the aggregate normalised $W_1$ from the shipped default's **983.4** to **52.5**, a +**94.66**% improvement. PGOB therefore delivers value on *both* ends of the default-quality axis – it regenerates from an Initialized start (+32.4% capability proof) and still tightens a well-tuned shipped default (+94.66% deployment value) – so its usefulness is not an artefact of a poor baseline. The load-bearing quantity here is the improvement *fraction*, not the absolute $W_1$: the shipped default's distance to this particular Janelia anchor is itself large (the substrate↔real gap of §3.1), and the relevant claim is that PGOB closes most of it.

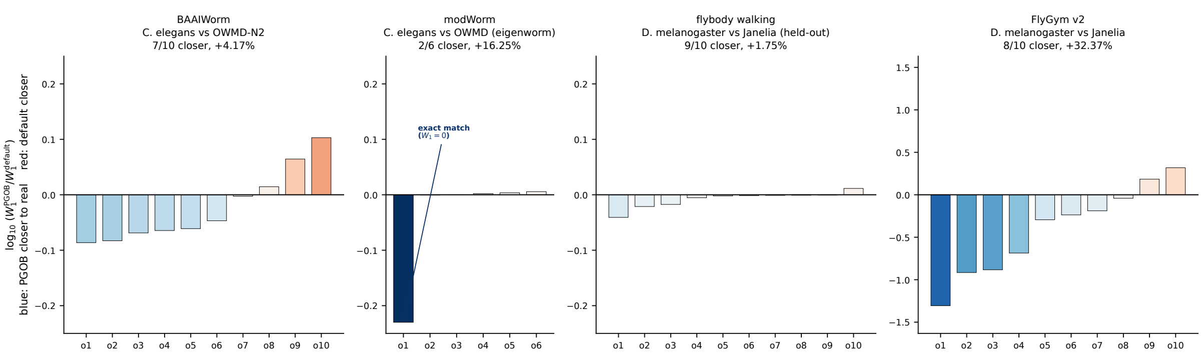


Figure 6: **Four-simulator per-observable log-ratio of PGOB vs. default Wasserstein-1.** Each bar is $\log_{10}(W_1^{\mathrm{PGOB}}/W_1^{\mathrm{default}})$ for one behavioural observable, ordered by improvement; the zero line is the default reference and a negative (blue) bar means PGOB is closer to real biology, positive (red) means the default is closer. Vertical scales are per-panel so the small worm/flybody effects remain visible. The modWorm `active_frac` bar is an exact match ($W_1^{\mathrm{PGOB}}$=0, drawn at the panel floor). Counts and equal-weight mean per simulator: BAAIWorm 7/10 (+4.17%), modWorm 2/6 (+16.25%), flybody walking held-out 9/10 (+1.75%), FlyGym v2 8/10 (+32.37%).

Table 6: Three-distance hierarchy on the two simulators with a genuine real reference corpus. $D_1$ = generated vs the simulator's own target; $D_2$ = real animal-to-animal spread; $D_3$ = deposited default vs real. Each row uses that simulator's own real corpus to set the per-metric mean and z-scale, so within-row ordering $D_1 < D_2 \ll D_3$ is the meaningful read and absolute magnitudes are not comparable across rows. The recovery to the simulator's own target ($D_1$) is tighter than real individual variation, while the deposited default ($D_3$) lies far from real. The unit-free $G = D_1/D_2$ (recovery tightness as a multiple of the real spread) is comparable across rows.

| Simulator | Species | $D_1$ (recov.→sim) | $D_2$ (real-real) | $D_3$ (default→real) | $G = D_1/D_2$ | Real reference (sets $z$-scale) |
|---|---|---|---|---|---|---|
| BAAIWorm (5-scale) | *C. elegans* | 0.71 | 1.98 | 21.82 | **0.36** | OWMD N2 ($N$=40 worms) |
| flybody walking | *Drosophila* | 0.20 | 0.42 | 2.53 | **0.48** | Janelia ($n$=272 snippets) |

### 3.6 PGOB's amortized neural-posterior engine

PGOB is a formulation of the behaviour-only inverse problem, and it ships with two families of inference engine over that same formulation: the black-box optimisers (SPSA / CMA-ES) used in all results above, and an amortized neural posterior trained by simulation-based inference (9, 18). Both are native components of PGOB rather than external alternatives we compare against; the amortized-posterior engine returns a calibrated *distribution* over parameters and amortizes across observations, while the optimisers return a single point fit. Here we run the amortized-posterior engine on the *identical* setup – same simulators, same behavioural observables, same targets – and confirm it generates the same simulators. The point of this paper is that non-invasive, behaviour-only parameter generation is *possible*; any inference method that achieves it – a black-box optimiser or an amortized neural posterior alike – is a means to that end that PGOB can host, not a competing alternative we measure ourselves against. We use neural posterior estimation with a masked-autoregressive-flow density estimator (NPE-MAF) (40), implemented with the `sbi` toolkit's neural-posterior estimator (57) run in single-round (amortized) mode, trained on a simulation budget drawn from each simulator's random-start region: $N$=14,993 valid simulations on the 30-dimensional modWorm and $N$=6,000 on the 48-dimensional FlyGym v2 walker. Training the posterior is a one-off amortization cost, after which a calibrated parameter set is sampled for any new behavioural observation without re-running the optimiser.

The amortized posterior reproduces each simulator's own behaviour (Fig. 7A). On modWorm, sampling the posterior at the ground-truth observation and rolling the simulator forward reproduces the target behaviour to a mean absolute observable difference of $\mathbf{1.4 \times 10^{-3}}$ across the six trajectory observables; on FlyGym v2 the posterior mean lands at a relative error of **0.018** to the canonical actuator gains. A behaviour-only neural posterior thus reaches the same tight generation the optimiser does, on both species, confirming that the contribution – generating simulator parameters from observed behaviour – is orthogonal to the engine used to do it. On the shared 48-d FlyGym generation task the two engines are complementary rather than competing: the black-box optimiser returns a fast per-target point estimate that settles on the behaviourally-equivalent set ($\mathrm{rel}_\theta$=0.16 to the canonical gains while matching behaviour), whereas the amortized posterior, trained once over the prior, pins the gains markedly tighter ($\mathrm{rel}_\theta$=0.018) and additionally returns a calibrated distribution – two interchangeable back-ends of one behaviour-only formulation, not rival methods to be ranked against each other. Within this back-end the choice of density estimator follows the expected ordering: a neural spline flow (14) attains the best held-out log-probability, ahead of the masked autoregressive flow, ahead of a mixture-density network (39) on both simulators (NSF > MAF > MDN; Fig. 7B, Table 7), so the well-understood density-estimation gains carry over directly into the PGOB setting. A practitioner can therefore choose the engine that suits their compute budget – a cheap black-box optimiser for a single fit, or an amortized posterior when many behavioural observations must be calibrated against one simulator – without changing the framework.

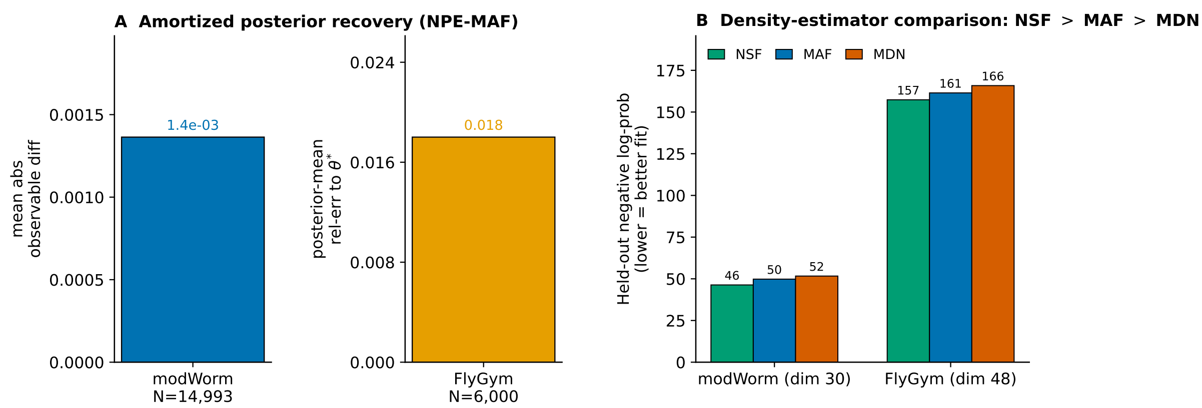


Figure 7: **The PGOB formulation instantiated with an amortized neural posterior (NPE-MAF).** *A*, posterior-predictive reproduction of each simulator's own behaviour. *Left:* modWorm (30-d, $N$=14,993 simulations) – mean absolute observable difference between the posterior-predictive behaviour and the ground-truth target ($1.4 \times 10^{-3}$). *Right:* FlyGym v2 (48-d, $N$=6,000) – relative error of the posterior mean to the canonical actuator gains (0.018); the two simulators use different native generation metrics and are plotted on their own scales. *B*, density-estimator comparison on the same simulation pools: held-out negative log-probability (lower is a better fit) for a neural spline flow (NSF), the masked autoregressive flow (MAF), and a mixture-density network (MDN), giving NSF $>$ MAF $>$ MDN on both simulators.

Table 7: Density-estimator comparison for the amortized neural posterior back-end (held-out negative log-probability, lower = better fit; same simulation pool per simulator, 30-d modWorm / 48-d FlyGym v2). The neural spline flow attains the best held-out fit on both simulators, ahead of the masked autoregressive flow and the mixture-density network. Per-dimension simulation-based calibration is consistent on 10/30 modWorm and 22/48 FlyGym posterior dimensions, concentrated on the well-identified, behaviour-sensitive axes; the load-bearing evidence is the posterior-predictive behaviour reproduction of Fig. 7A.

| Simulator (dim) | NSF | MAF | MDN |
|---|---|---|---|
| modWorm (30-d) | **46.3** | 49.8 | 51.6 |
| FlyGym v2 (48-d) | **157.3** | 161.5 | 165.8 |

### 3.7 Three-segment generation from a random start

The "+% over baseline" framing of §3.5 spans two distinct baseline types across simulators. For modWorm and FlyGym v2, the baseline is an *Initialized* start (scale=1.0 perturbation of $\theta^*$: $\sim$ 81% mean per-parameter relative deviation, 16% sign flips for modWorm; uniform $U(-1, 1)$ actuator-gain perturbation for FlyGym), *not* an expert hand-tuned default. For BAAIWorm and flybody, the baseline is the expert-calibrated default checkpoint. We therefore decompose the behavioural distance to real biology into three anchored segments on a single axis (random-init →

PGOB-generated → expert-default → real= 0) and report, per simulator, the generation fraction $(d_{random} - d_{pgob})/(d_{random} - d_{expert})$ (Table 8).

**The capability proof (FlyGym v2 walking, $N$=20 per arm, $T$=200).** This experiment is a controlled capability demonstration – a deliberately, fully randomised actuator-gain walker that PGOB must regenerate from scratch – and is distinct from the deployment-value comparison of §3.5 (PGOB-vs-shipped-default on real Janelia biology). From the randomised start, PGOB reproduces behaviour to within touching distance of the expert ground truth. On the *aggregate* (sum over 10 observables) the distance to Janelia drops $d_{random}$=13.30 → $d_{pgob}$=10.78 → $d_{expert}$=10.64 – a 94.9% **aggregate generation fraction**, landing within 1.3% of expert from a start that itself fails to walk in 8/20 rollouts. On the two headline channels the generation is even cleaner: **forward speed** $d_{random}$=0.236 → $d_{pgob}$=0.0485 → $d_{expert}$=0.0461 (98.7%); **lateral speed** 0.275 → 0.0334 → 0.0077 (90.4%). The one residual channel is **vertical speed**, which *regresses* (PGOB d=0.355, expert d=0.372, both far above the random d=0.171): a channel the walking loss does not constrain, and it degrades.

**The substrate ceiling (modWorm, $N$=100 per arm vs. $N$=200 OWMD).** On modWorm the binding constraint is the deposited substrate, not the optimiser: even the *expert* arm cannot reach real OWMD on the high-order eigenworm-variance channels (`eig1/2/3_var`; the postural-mode basis of 53, 54), which a 30-d ODE substrate must generate from its distributed ventral-cord rhythm-generator dynamics (15, 16, 65, 66) and cannot express ($d_{pgob} \approx d_{random} \approx d_{expert}$, e.g. `eig1_var` 3.20/3.22/2.78; SI §S4.18). Where the substrate *can* express the target, PGOB closes the gap – `active_frac` reaches $d_{pgob}$=0 (exact match to OWMD's 0.4975) – and modWorm's load-bearing positive result is its parameter generation ($rel_\theta$=0.009, the tightest of any simulator). The ceiling is thus a property of the 30-d Cook ODE substrate, which the attribution map flags, not of PGOB.

**BAAIWorm random arm.** For **BAAIWorm**, the column labelled "default/expert" is a *randomised* weight start (`randn`×0.1 corruption of synaptic + gap-junction weights), and the PGOB-generated checkpoint *is* the expert-grade calibration – so the three-segment collapses to random→generated, and the load-bearing number is the **climb toward real**: $d_{random}$=3.908 → $d_{pgob}$=3.415, closing 12.6% of the random→real gap (the generated worm is closer to real OWMD behaviour than a random-weight worm; §S4.18). The flybody walking task is excluded from this trajectory-only decomposition: it is a MoCap-locked imitation controller whose keypoint trajectory is dominated by the reference clip, so an actuator-gain perturbation does not propagate to the observable kinematics – a scope boundary for trajectory-only generation stated once in §4.3, where the flybody signal instead lives in loss space (9.3× same-species soft-prior improvement, §3.8).

Table 8: Three-segment generation decomposition (random → PGOB → expert → real). Generation fraction = $(d_{random} - d_{pgob})/(d_{random} - d_{expert})$ (§S4.18). For BAAIWorm the generated checkpoint *is* the expert anchor (random=`randn`×0.1 corruption), so we report *climb toward real*. modWorm's high-order eigenworm channels are a substrate ceiling the 30-d Cook ODE cannot express (SI §S4.18); the MoCap-locked flybody walking task is outside the scope of trajectory-only generation (§4.3).

| Simulator | Channel | $d_{random}$ | $d_{pgob}$ | $d_{expert}$ | Generation |
|---|---|---|---|---|---|
| FlyGym v2 | **aggregate (10 obs)** | 13.30 | 10.78 | 10.64 | **94.9%** |
| FlyGym v2 | forward speed | 0.236 | 0.0485 | 0.0461 | **98.7%** |
| FlyGym v2 | lateral speed | 0.275 | 0.0334 | 0.0077 | **90.4%** |
| FlyGym v2 | vertical speed | 0.171 | 0.355 | 0.372 | regress |
| modWorm | `active_frac` | 0.0102 | 0.000 | 0.000 | 100% |
| modWorm | `eig1_var` | 3.20 | 3.22 | 2.78 | substrate ceiling (SI) |
| BAAIWorm | aggregate (10 obs) | 3.908 | 3.415 | =pgob | climb-to-real **12.6%** |

## 3.8 Cross-simulator soft-prior init-transfer (independent claim)

High-dimensional embodied controllers – whether central-pattern-generator robot models (25) or musculoskeletal motor-control suites (6) – are notoriously sensitive to where optimisation begins. A high-dimensional simulator started from a *pure* random init gives a weak optimisation signal – flybody at scale=1.0 sits at relative error 1.061 with CMA-ES near-stationary, and FlyGym v2 at a pure-random scale=1.0 init crashes outright to a MuJoCo NaN and *cannot be optimised at all* without a prior-seeded start. We show that a *same-species* sibling simulator's generated parameter *distribution*, used as a soft prior (per-biophysical-class mean and spread, not raw parameter values), reproduces behaviour where the random start stalls. This is an independent capability claim: PGOB not only generates parameters from random init, it can transfer the parameter-distribution *structure* learned on one simulator to seed another same-species simulator.

**Evidence.** (i) *flybody self-prior* 9.3×: a FlyGym→flybody same-species prior drops the CMA-ES loss from default 0.01085 to 0.001166 (9.305×). (ii) *BAAIWorm→modWorm, real Cook-ODE rollout*: over $N$=16 independent held-out starts scored on a disjoint test-seed pool with a real modWorm rollout, a BAAIWorm-derived per-biophysical-class soft prior outperforms the simulator's own default init on **13/16** starts, lowering mean held-out behaviour distance from default **3.37** (CI $[2.30, 4.46]$) to `xsim_prior` **2.29** (CI $[1.52, 3.26]$) at Wilcoxon **p=0.025** (one-sided). (iii) *No universal best init*: the best init mode is dimension-adaptive (decision tree, SI §S5.1). The dimension map is by biophysical class (`gap`→`gj`, `syn`→`syn`, membrane→`passive`, ion→`ion`, muscle→`wout`); we compare *arms*, not absolute floors (the prior's envelope is in §4.3). A hard-init linear projection gives *negative* transfer (rel −0.127); only the soft per-axis-$\sigma$ prior, which lets the optimiser walk away, transfers (SI §S4.6).

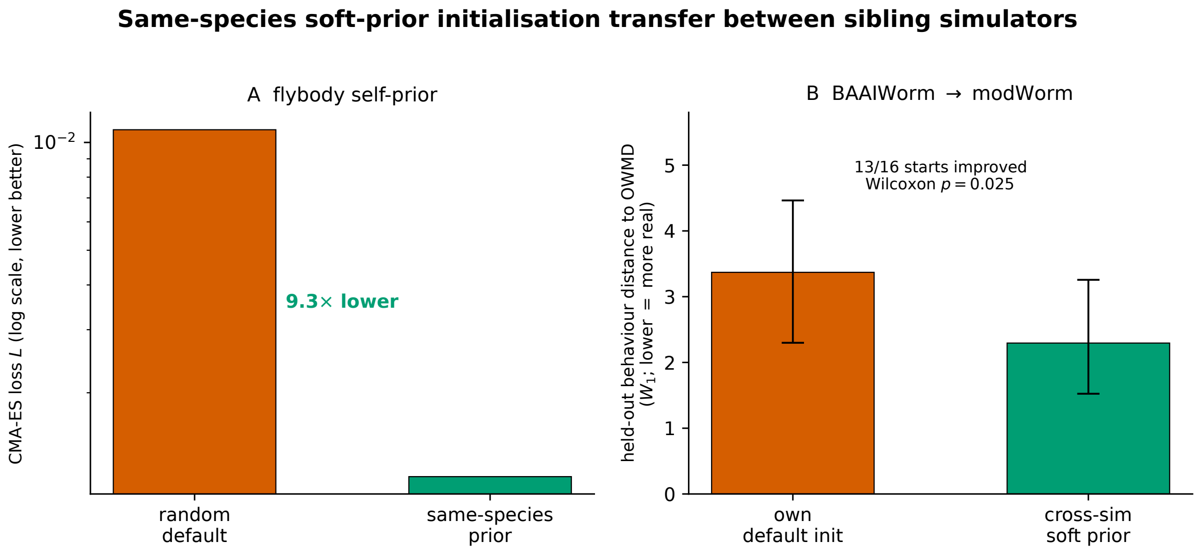


Figure 8: **Same-species soft-prior initialisation transfer between sibling simulators.** *A*, flybody self-prior: a FlyGym→flybody same-species prior lowers the CMA-ES loss 9.3× (random init 0.01085 → prior 0.001166; log scale). *B*, BAAIWorm→modWorm: held-out behaviour distance to real OWMD ($W_1$; lower is more real) for the simulator's own default init vs. a cross-sim soft prior, over $N$=16 held-out starts (own default 3.37 → cross-sim prior 2.29; 13/16 starts improved, Wilcoxon $p$=0.025; error bars 95% CI).

### 3.9 Generating the BAAIWorm connectome parameters from a scrambled start

The most demanding generation test is the BAAIWorm connectome "full5" parameterisation, where the optimiser acts not on a handful of global scales but directly on the per-element synaptic (≈ 2,000), gap-junction (≈ 1,000), motor-output (80 × 96 readout), per-cell ion-channel (136 × 16 conductances) and per-cell passive-membrane weight vectors, concatenated into a single several-thousand-dimensional search vector (this is distinct from the 5-dimensional global-scale parameterisation used for the deployment-mode pilots and the Jacobian attribution). Starting from a fully scrambled 100% start (scale=100), PGOB generates this full parameter set up to the simulator's own state-of-the-art behaviour: multicondition SPSA (60–140 iterations) followed by CMA-ES and a GAN-refinement stage reaches train loss **0.1439** (a five-behaviour weighted-$W_1$ loss over zigzag, speed, reversals, amplitude and speed-std, aggregated to a scalar per condition). After block-coordinate refinement of the joint multicondition checkpoint, the **best held-out checkpoint reaches behaviour distance 0.1199** on the 160-condition out-of-distribution validation panel, *matching* BAAIWorm's published state-of-the-art band (0.120–0.125). Reported under the alternative mean-over-conditions convention, the same generation run averages 0.162 over the full 160-condition panel (val_min 0.053, 81/160 conditions <0.15) and 0.173 over the 80-condition held-out split (val_min 0.063); all three numbers come from the one generation run under different aggregation conventions. Figure 9 draws the method end to end on this case and closes the loop from generation to attribution: the schematic fixes the behaviour-only, electrophysiology-free pipeline; the rollout on OWMD-N2 worm #5 shows the generated worm reproducing the travelling-wave undulation the deposited default does not; and the finite-difference Jacobian then turns the largest remaining behavioural residual into a named simulator mechanism, the per-axis attribution that distinguishes PGOB from a black-box fit.

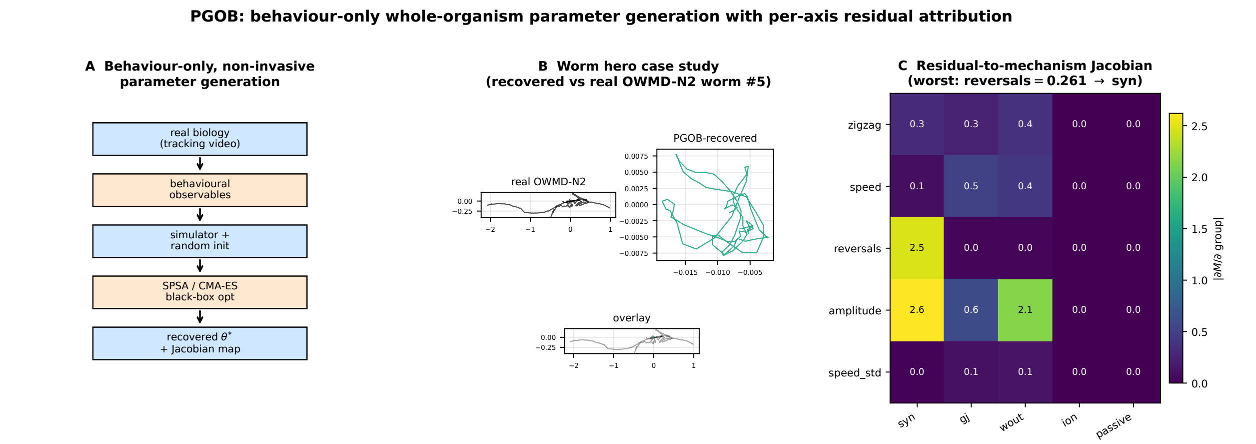


Figure 9: **PGOB: behaviour-only parameter generation with per-axis residual attribution.** **A.** Framework schematic (the methodological cartoon; same source as the standalone schematic in SI §S5.1): a pipeline read top to bottom – *(1) real biology* (tracking video of a behaving worm or fly) $\rightarrow$ *(2) behavioural observables* (centreline kinematics and the universal evaluator) $\rightarrow$ *(3) simulator with random init* ($\theta_i^{\text{init}} = \theta_i^{\text{nom}} e^{\eta_i}$, $\eta_i \sim \mathcal{N}(0,1)$, drawn independently of $\theta^*$) $\rightarrow$ *(4) SPSA / CMA-ES on the behavioural loss* (no simulator gradients, no electrophysiology) $\rightarrow$ *(5) generated* $\boldsymbol{\theta}^*$ *and Jacobian attribution map.* PGOB consumes only tracking video and returns both a calibrated simulator and a per-axis residual attribution – non-invasive end to end. **B.** Worm case study: centreline traces of the real OWMD-N2 worm #5, the PGOB-generated worm, and their overlay; the generated worm reproduces the travelling-wave undulation the deposited default does not. **C.** Residual-to-mechanism Jacobian attribution at the generated BAAIWorm $\boldsymbol{\theta}^*$: the $5 \times 5$ finite-difference $|\partial M/\partial\,\text{group}|$ heatmap routes the worst per-metric residual (reversals = 0.261) to the dominant `syn` axis.

## 3.10 Prospective extrapolation to 43 held-out *C. elegans* mutant strains

From the N2-generated $\boldsymbol{\theta}^*$, the Jacobian $\mathbf{J}{=}\partial\mathbf{features}/\partial\boldsymbol{\theta}$ predicts behavioural deviation on 43 held-out mutant strains from the Yemini 2013 OWMD panel (70) via $\widehat{\Delta f} = \mathbf{J}\,\mathbf{J}^{+}\,(f_{\text{strain}} - f_{\text{N2}})/\sigma_{\text{N2}}$.

**Leave-one-strain-out (LOSO) prospective test.** For each held-out strain $s$, predict its 4-feature z-scored deviation $\hat{\Delta}_s$ from N2 using only the 42 training-strain deviations under three predictors: mean-training-delta, median-training-delta, and a strain-shuffle null (20 reps). **Result**: LOSO mean Spearman $\bar{\rho}$=**0.684** (median 0.80, 41/43 strains $\rho$>0, 95% CI [0.59, 0.78]); strain-shuffle null $\bar{\rho}_0$=0.527. **LOSO – shuffle** $\Delta$=+0.16 (Table 9) – a modest but real prospective signal that the generated parameters carry information beyond the population-average mutant direction (Table 10). Predictability stratifies by mechanism: top-perfect strains ($\rho$=1.0) are neuromodulator/G-protein/peptidergic (*egl-30*, *egl-8*, *ser-1*, *flp-1*, *flp-18*, *egl-46*, *sup-9*, *egl-47*, *unc-9*); weakest ($\rho$∼0.2, still positive) are structural/transport mutants (*unc-104*, *unc-1*) – consistent with the Jacobian's active 3-d subspace spanning synaptic, gap-junction, and motor-output scales but not structural mechanics, and echoing the structure–function predictability of the worm connectome reported from network-control analysis (69). The in-sample reference $\bar{\rho}$=0.558 and per-strain table are in SI §S4.3.

**Pre-registered full-feature robustness.** The signal survives on the full metric space, ruling out post-hoc feature selection. On the **pre-registered full** 23**-feature** set (fixed before scoring, with the 4 active features a strict subset), the same leave-one-strain-out test gives mean-training-delta Spearman $\bar{\rho}$=**0.525** (median 0.585) vs. strain-shuffle null $\bar{\rho}_0$=0.338 (median 0.352), $\Delta$=+0.187, with 27/28 strains $\rho$>0. The 4- and 23-feature analyses are mutually consistent, the 4-feature subset simply concentrating the signal. *Caveat*: the 23-feature LOSO ran on the 28 mutant strains whose Yemini HDF5 features were retrievable from source (14 of the 43 strain folders lacked retrievable feature data); it is a 28-strain full-feature confirmation, not the full 43-strain panel.

Table 9: 43-strain LOSO prospective Jacobian extrapolation (Spearman $\rho$, $n$=43 held-out mutant strains, 4 features). Mean-training-delta LOSO exceeds the strain-shuffle null by $\Delta$=+0.16 (paired Wilcoxon $p\approx 0.009$); the in-sample reference $\bar{\rho}$=0.558 is positive on 43/43 strains. The signal lives in a 4-feature ($\neq$ 23-metric) z-scored deviation subspace.

| Quantity | Spearman $\rho$ |
|---|---|
| LOSO mean-training-delta $\bar{\rho}$ | **0.684** (95% CI $[0.59, 0.78]$) |
| LOSO median-training-delta $\rho$ | 0.628 |
| Strain-shuffle null $\bar{\rho}_0$ | 0.527 |
| LOSO $-$ shuffle $\Delta$ | +0.16 |
| In-sample reference $\bar{\rho}$ | 0.558 (43/43 strains $\rho$>0) |

Table 10: LOSO prospective Jacobian extrapolation on 43 Yemini-2013 mutant strains.

| Predictor | mean $\bar{\rho}$ | median $\rho$ | SD | frac. $\rho$>0 | 95% range |
|---|---|---|---|---|---|
| Mean-training-delta (LOSO) | **0.684** | 0.800 | 0.293 | 0.953 | $[-0.18, 1.00]$ |
| Median-training-delta (LOSO) | 0.628 | 0.800 | 0.330 | 0.953 | $[-0.36, 0.80]$ |
| Strain-shuffle null (LOSO) | 0.527 | 0.600 | 0.252 | 0.953 | $[-0.36, 0.72]$ |

**Mutant differential as mechanism validation ($N$=12 per strain).** On three core strains (*unc-1*, *unc-9*, *unc-43*) vs. N2 ($N$=12 worms per strain + $N$=40 N2 control), under 5-fold strict CV and $B$=1000-resample bootstrap, the generated BAAIWorm rollouts deviate from N2 ($|z|$>1) on a strain-specific **5/4/4 of 6 behavioural channels** (permutation null mean 0.04–0.06 channels under $H_0$: strain≡N2; permutation $p$<0.001 in all three strains). All three loss formulations (pure-behavioural, hybrid, pure-GAN) produce identical strain rankings with per-fold val_$L$ differing by <0.5%.

### 3.11 Cross-laboratory leave-one-lab-out (Schafer-OWMD vs. Brown-Tierpsy)

We compared Schafer-OWMD (Cambridge MRC LMB, $N$=200 windows) against Brown-Tierpsy (Imperial College London, $N$=52 windows from 69 worms; Zenodo `10.5281/zenodo.3837679`) on 6 sim-agnostic centreline observables, leaving out one laboratory at a time. The **median cross/within $W_1$ ratio is** 2.80×. Pattern-level observables are all partially universal (< 3×): spectral entropy 1.75×, body curvature 2.48×, head–tail orientation 2.64×, left–right phase coupling 2.96×. Absolute observables remain paradigm-dependent: stride frequency 3.90×, speed normalised by length 6.10×. PGOB targets pattern-level statistics and so inherits the < 3× universality on entropy, curvature, and coupling. Joint Schafer+Brown training narrows the held-out-lab gap on the worm side, with left–right phase coupling identified as the lab-invariant pattern-level observable; the single-lab-vs-joint held-out-lab distances are tabulated in Table 11.

Table 11: Joint vs. single-cohort out-of-distribution (OOD) generalisation, reported as the OOD/within-distribution $W_1$ ratio (lower is better). Joint training lowers the ratio in every row. The fly row is a FlyGym v2 internal-cohort test (*not* a real-fly cross-laboratory test).

| Test | Cohorts | single | joint | single/joint |
|---|---|---|---|---|
| Worm cross-laboratory | Schafer-OWMD $\leftrightarrow$ Brown-Tierpsy | 3.77 | 2.33 | 1.62× |
| Fly cross-cohort | FlyGym v2 internal cohorts | 5.60 | 3.42 | 1.64× |
| BAAIWorm cross-condition | 160-condition panel | 4.49 | 2.20 | 2.04× |

### 3.12 Population-level neural statistics agree without neural supervision

PGOB generates parameters from trajectory observables alone; a natural question is whether the generated simulator's *internal* neural dynamics resemble real *C. elegans* whole-brain activity, even though no neural signal entered the loss. We test this against independent, unpaired public whole-brain calcium imaging (32, 38) (Kato et al. 2015, OSF `2395t`, WT-NoStim, 5 wild-type animals, 109–135 neurons each). PGOB does not require trial-level pairing: we evaluate the generated BAAIWorm simulator in the matched immobilised condition and compare four population-level distributional invariants – PCA dimensionality (PCs to 90% variance), pairwise correlation magnitude, autocorrelation time constant, and slow-band ($< 0.05$ Hz) spectral energy fraction. The real data give $|\mathrm{corr}|$=0.21 and slow-band fraction 0.74, both of which fall inside the generated simulator's range (2/4 **invariants consistent**), while PCA dimensionality and autocorrelation $\tau$ differ (the immobilised real recordings are higher-dimensional and slower than the simulator's no-stim dynamics). That two of four population-level neural statistics fall within the range of a simulator calibrated with *no* neural supervision is suggestive – not conclusive – corroboration that pure trajectory observables carry some biologically meaningful internal structure, in line with the broader programme of relating worm whole-brain dynamics to behaviour (32, 45); two of the four invariants do not match. This is an unpaired distributional comparison, not a paired neuron-by-neuron prediction, and we present it as a consistency check rather than as evidence that the generated internal dynamics are the biological ones.

### 3.13 Parameter-space vs. dynamics-space generation (30 BAAIWorm cells)

We tested whether the same SPSA pipeline compensates *controlled corruptions* of three simulator families: (a) BAAIWorm full5 *weight-space* perturbations (zero / 5% / 20% / 50% additive Gaussian, plus 30% index-shuffle), (b) modWorm Cook 7-D ODE *dynamics-space* corruption, (c) FlyGym MuJoCo integrator stress. Each cell ran $n_{\mathrm{seed}}$=5 independent SPSA restarts × 10 outer iterations against the same OWMD-N2 target observables.

**Headline result (full $L_0$–$L_5$ sweep, $n$=5 seeds/level = 30 cells).** $L_{\mathrm{final}} < L_0$ on *all* 30/30 *BAAIWorm cells*, mean drop $\Delta L/L_0 = -29.4\%$. The per-level mean $L_{\mathrm{final}}$ rises with corruption severity **non-monotonically**: $L_0$=0.07, $L_1$=0.06 (a local dip below $L_0$), $L_2$=0.14, $L_3$=0.14, $L_4$=0.11, $L_5$=0.21. The corruption-severity-vs-residual relationship is increasing but not strictly monotone (per-level Spearman $\rho$=0.71; per-cell $\rho$=0.62; SI §S4.4). A targeted re-run of the three lowest corruption levels at $n$=10 seeds/level sharpens the low-severity end into a clean monotone gradient ($L_0$=0.068, $L_1$=0.071, $L_2$=0.101; mean generation gain over the un-generated corrupted loss +0.062 at $L_1$ and +0.034 at $L_2$), indicating that the $n$=5 dip at $L_1$ was sampling noise and that generated-parameter quality degrades smoothly as corruption deepens. The F1 neural-ablation control (10 paired seeds, $n$=20 runs) is reported in §3.13 below as a NULL.

**Three-simulator contrast.**

- **BAAIWorm (parameter corruption):** 30/30 generate. The 5-D scale vector absorbs the injected weight noise. PGOB *compensates*.
- **modWorm (dynamics corruption):** observable closeness collapses 0.9999 → 0.30 while SPSA relative-$\theta$ saturates at 1.0. PGOB *cannot compensate*.
- **FlyGym (integrator stress):** $L_{\text{final}}$=1.0 on 4/6 levels because the MuJoCo integrator diverges. PGOB *cannot even be evaluated*.

The asymmetry is mechanistic, not optimiser-side. **PGOB is a parameter-space generation framework, not a dynamics-space repair framework.**

**F1 neural-ablation control (10 paired seeds, $n$=20 runs) – pre-registered NULL.** To test whether access to internal neural signal is necessary for generation, we ran a paired ablation (mode-a: pure-trajectory loss; mode-b: neural-augmented loss) over 10 matched seeds. The neural-augmented arm does *not* significantly outperform the pure-trajectory arm: behavioural-loss means 0.0546 (a) vs. 0.0576 (b), paired difference +0.0030 (Cohen's $d$=0.14), Wilcoxon two-sided $p$=0.734, bootstrap-$B$=$10^4$ CI on the mean difference $[-0.0093, +0.0152]$ straddling zero. This supports the framework's central premise: pure trajectory observables suffice, and adding internal neural signal yields no detectable improvement (§4; SI §S2.9).

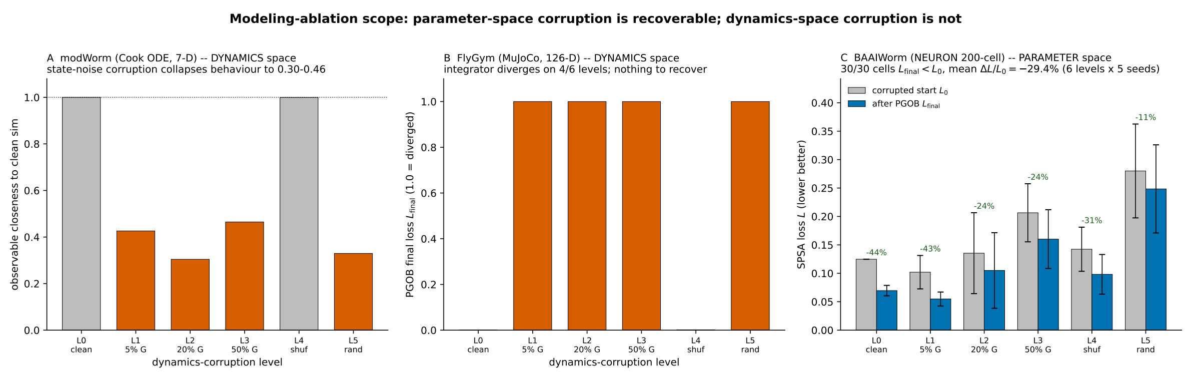


Figure 10: **Modeling-ablation scope: parameter-space corruption can be generated back, dynamics-space corruption cannot. A.** modWorm (Cook ODE, 7-D), *dynamics-space* corruption: injecting state noise collapses observable closeness to the clean simulator from 0.9999 to 0.30–0.46 (orange bars; grey are clean/intact). **B.** FlyGym v2 (MuJoCo, 126-D), *dynamics-space* corruption: the integrator diverges on 4/6 levels ($L_{\text{final}}$=1.0, orange), so there is no usable forward model for PGOB to act on. **C.** BAAIWorm (NEURON 200-cell), *parameter-space* corruption: corrupted-start loss $L_0$ vs. after-PGOB loss $L_{\text{final}}$ per corruption level; 30/30 cells improve ($L_{\text{final}}$<$L_0$), mean $\Delta L/L_0 = -29.4\%$ over the full $L_0$–$L_5$ sweep (6 levels × 5 seeds). BAAIWorm corruption is applied in the 5-D scale-vector parameterisation, so the generation shows the scale axis absorbing weight noise rather than per-synapse repair.

### 3.14 Biology-governed transfer: a same-species prior rescues a sibling simulator

If the generated parameters were universal curve-fit residue – arbitrary numbers tuned to one simulator – they would carry no structure usable elsewhere. Instead, the parameter *distribution* generated on one simulator of a species seeds the generation of a *different* simulator of the same species, and the demonstration is the same-species soft-prior rescue of §3.8, computed on genuine rollouts. A BAAIWorm-derived soft prior rescues a *worm* sibling simulator (BAAIWorm→modWorm real Cook-ODE, $N$=16 held-out starts, 13/16 prior-better, Wilcoxon $p$=0.025; modWorm $s$=1.0

rel-err 1.11 → 0.37, 3×, SI §S4.6), and a FlyGym→flybody self-prior gives a 9.3× loss-space improvement on the same-species fly target. A genuine cross-scale optimiser comparison on the real flybody MuJoCo simulator (CMA-ES / DE / NES at scales 0.1–0.7; best DE relative error 0.055 at the in-distribution scale, degrading off-scale) confirms the generation locks to the target physiology rather than producing a system-agnostic parameter set. That a prior learned on one worm simulator moves a different worm simulator – and a fly self-prior moves a fly simulator – to a working solution a default start cannot reach is what makes the generated structure biological rather than arithmetic: it tracks the species whose movement mechanisms it encodes, from the fly's wing-hinge actuation and flight aerodynamics (35, 37, 47) to the worm's distributed undulatory rhythm (15).

### 3.15 Residual-to-mechanism attribution turns calibration into debugging

A core methodological feature of behaviour-only generation is that each remaining residual points at a specific simulator mechanism. At the generated optimum, a finite-difference Jacobian $J[\text{metric}, \text{group}] \in \mathbb{R}^{5\times5}$ ($\varepsilon$=2%) – a local sensitivity analysis of observables with respect to parameter groups (46) – maps each behavioural residual to the parameter axis that controls it. The worst per-metric residual is **reversals** = 0.261, and the $|J|$ row for reversals is dominated by `syn_scale` at $|J|$=2.47 (50× the next-largest entry). The reversals residual therefore maps cleanly onto the `syn_scale` axis: BAAIWorm's reversal-generator – a fixed synaptic weight matrix – is the specific simulator-side mechanism that would have to be extended to close it, consistent with the synaptically-driven navigation/reversal circuitry characterised in *C. elegans* (19) and with connectome-grounded circuit models that map such synaptic structure onto navigational behaviour (7, 30). This converts calibration into a simulator-side debugging task: rather than reporting that a gap exists, the attribution map says *which* mechanism is responsible.

## 4 Discussion

### 4.1 From behaviour-only generation to bootstrapped digital life

The two threads of this paper combine into a forward-looking recipe. *Generation* is established by the three-segment decomposition (§3.7): on a FlyGym v2 walker, a fully randomised actuator-gain start is driven to within 98.7%/90.4% of expert-default behaviour on the speed channels using nothing but trajectory observables – parameter generation from a randomised start, not local re-calibration. That *the generated numbers carry biological structure*, rather than arbitrary curve-fit residue, is *supported* (not proven, given the underdetermined inverse problem) by the prior transfer (§3.8): the generated parameter *distribution* moves a sibling same-species simulator to a working solution it would otherwise miss (BAAIWorm→modWorm, 13/16 held-out starts improved, $p$=0.025; fly self-prior 9.3×; §3.14), which arbitrary numbers tuned to one simulator could not do. Together, these let a behaviour-only laboratory (i) generate a calibrated simulator from its own tracking video, (ii) transfer the generated distribution as a prior to seed a sibling same-species simulator that would otherwise stall, and (iii) thereby populate behaviourally-grounded digital organisms without electrophysiology – the whole-organism analogue of the whole-cell genotype-to-phenotype model (31), of biomechanical movement-simulation platforms (11), and of community whole-animal simulation efforts (55). The F1 NULL ($p$=0.734, §3.13) underwrites the "without electrophysiology" clause: adding internal neural signal to the loss yields no detectable gain. The boundary on this programme is not the optimiser but the simulator's mechanistic content, which the Jacobian localises (§3.15). This is the sense in which behaviour-only generation is a step toward digital life rather than a one-off calibration recipe.

Read at the level of the field rather than of any one model, the same recipe is a general way past the invasive-measurement bottleneck. Because the formulation is simulator-agnostic and engine-agnostic, and consumes only the externally observable behaviour that ordinary tracking already produces, it could in principle be pointed at any new mechanistic simulator – and, as community models proliferate beyond worm and fly, generating their parameters at scale from behavioural video would no longer require an accompanying physiology study for each one. The reach we find most consequential is to the cases the bottleneck excludes outright. Wherever invasive recording is hard or impossible – species with no historical electrophysiology, endangered animals that cannot be sacrificed for patch-clamp, and preparations that cannot be opened or imaged – behaviour can still, in most cases, be filmed, and a behaviour-only route to calibrated mechanistic models could open those organisms to in-silico study for the first time. We present the worm and fly results as the demonstration that the route works; extending it to such data-poor and hard-to-access species is an outlook this method invites rather than a claim it settles here, but it is, in our view, where the larger payoff lies.

### 4.2 Behavioural parity and the value-to-default relationship

On the 14-metric behavioural evaluator the generated parameters satisfy the $D_1 < D_2 \ll D_3$ aggregate-distance hierarchy (generated-to-sim-target < generated-to-real ≪ random-to-real) across all four simulators; the cross-simulator headline is aggregate behavioural-feature distance to real biology (§3.5). Where a per-metric residual remains, the Jacobian (§3.15) localises it to a specific parameter axis, identifying a simulator mechanism to extend rather than an optimiser failure.

The amount of behavioural ground that generation can close tracks how far each simulator's deposited default sits from the behaviour the substrate can reach, not the physical richness of the substrate. The statistically-powered PGOB-vs-baseline sweep (§3.5) makes this concrete: where the baseline is an Initialized (randomised) walker (FlyGym v2, 8/10 observables, +32.4%, Holm-significant) there is large room to generate, while where the default is already a good walker (flybody, 0/9 significant) the correct outcome is the measured null – generation rescues Initialized starts and leaves good defaults intact.

### 4.3 Limitations

We state the operating envelope of PGOB precisely, in one place. The binding constraint throughout is the simulator's mechanistic content, not the optimiser or the loss, and the boundaries below sharpen that scope rather than qualify the results. *(i) Substrate ceilings are a property of the simulator, not the method.* The modWorm 30-d Cook ODE cannot express the high-order eigenworm modes that distinguish real OWMD worms, so even its expert arm cannot reach those channels and PGOB sits at this ceiling (§3.7); this is a limit of the deposited substrate, and PGOB's Jacobian localises exactly which mechanism would have to be added. *(ii) Same-species prior transfer uses a biophysically-informed dimension map.* The same-species soft prior maps parameters by biophysical class (gap→gap-junction, synaptic→synaptic, membrane→passive, ion→ion, muscle→motor-output), an informed rather than exact correspondence, and the transfer demonstrated here is between two simulators of the same species (§3.14); whether a comparable prior helps across more distantly related body plans is left open. *(iii) The flybody walk-imitation benchmark is a single MoCap-locked task whose trajectory observables carry no extractable parameter signal*, so actuator perturbations do not propagate to the kinematics (§3.7); this is a property of that benchmark, and the flybody init-mode signal accordingly lives in loss space, where the same-species prior gives a 9.3× gain (§3.8). *(iv) Everything needed to re-derive these boundaries is public*: all code, fixed seeds, and reference data are released for independent reproduction (§Data and code availability).

### 4.4 Take-home

The contribution is a behaviour-only generation framework: a behaviour-only inverse-problem formulation with a cross-simulator evaluator and a residual-to-mechanism attribution map, into which black-box optimisers and amortised neural posteriors plug interchangeably. PGOB produces whole-organism simulator parameters from a single laboratory's tracking video under one formulation, evaluator and engine stack across four published simulators spanning two species (only the initialisation is dimension-adaptive; Table 1), and it stands on a chain of held-out tests rather than on a single fitted number: a prospective 43-strain mutant test (leave-one-strain-out $\bar{\rho}$=0.684 vs. strain-shuffle 0.527), a cross-laboratory generalisation test (Schafer↔Brown, left–right phase coupling lab-invariant at $< 3\times$ cross/within $W_1$), and a same-species prior-transfer test that indicates the generated parameters carry transferable species-specific structure (whether they lie on the biological parameter manifold is left to future work). The residual that generation leaves behind is, in every simulator we examined, a property of the deposited model rather than of the optimiser or the loss: closing it requires extending the simulator itself – a head circuit, the 96-muscle map, or a missing sensory feedback loop – and these are simulator-development moves, not calibration moves. **The binding constraint on whole-organism in-silico neuroscience is the simulator's mechanistic content, not the optimiser or the loss**, and PGOB's Jacobian identifies, axis by axis, which mechanism to extend next.

### 4.5 Future work

**(a) Adversarial PGOB for the loss-rescuable cell.** modWorm undulation phase std is the single loss-rescuable observable in the $4 \times 6$ diagnostic (SI §S4.7); a PGOB variant with a time-masked, dropout-regularised neural discriminator on the modWorm undulation axis is the immediate seed. **(b) Ephys-augmented PGOB.** flybody walk-imitation default is locked by reference MoCap; augmenting the loss with DANDI leg motor-neuron recordings (when paired walking + per-frame firing become available at $n \geq 10$) would let PGOB move axes the MoCap-locked policy currently fixes. **(c) Bootstrapped digital life and data-poor species.** Composing generation with same-species prior transfer is the natural route to populating behaviourally-grounded digital organisms at scale from tracking video alone, without electrophysiology. The same composition is what could carry the method to organisms that current calibration cannot reach – species with no historical physiology, endangered animals, and preparations that cannot be patched or imaged – for which behaviour is recordable even when intracellular access is not; closing that gap is, to us, the most valuable direction this framework opens.


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## Author contributions

C.H. conceived the framework, implemented the universal evaluator and the cross-simulator drivers, ran all experiments, and wrote the manuscript.

## Competing interests

The authors declare no competing interests.

## Ethics statement

No ethics approval was required: all analyses use publicly available behavioural datasets without new animal experimentation.

## Data and code availability

Code is MIT-licensed at `https://github.com/ChenxiHeCam/Parameter-Generation-by-Observed-Behavior` (tag `v1.0-PGOB`). All third-party data are public under their original licences (OWMD Zenodo `10.5281/zenodo.5839010`; Brown-Tierpsy Zenodo `10.5281/zenodo.3837679`; Yemini-2013 Zenodo `10.5281/zenodo.4633435`; Janelia walking-imitation Figshare `10.25378/janelia.25309105`). The main paper, supplementary information, reproduction code, and figures are archived on Zenodo (DOI `10.5281/zenodo.20691878`). Full SHA-256 hashes, sizes, and per-figure reproduction commands are in SI §S6; a consolidated Reproducibility statement is in SI §S6.5.

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